

INOS 6221 Assignment Guide

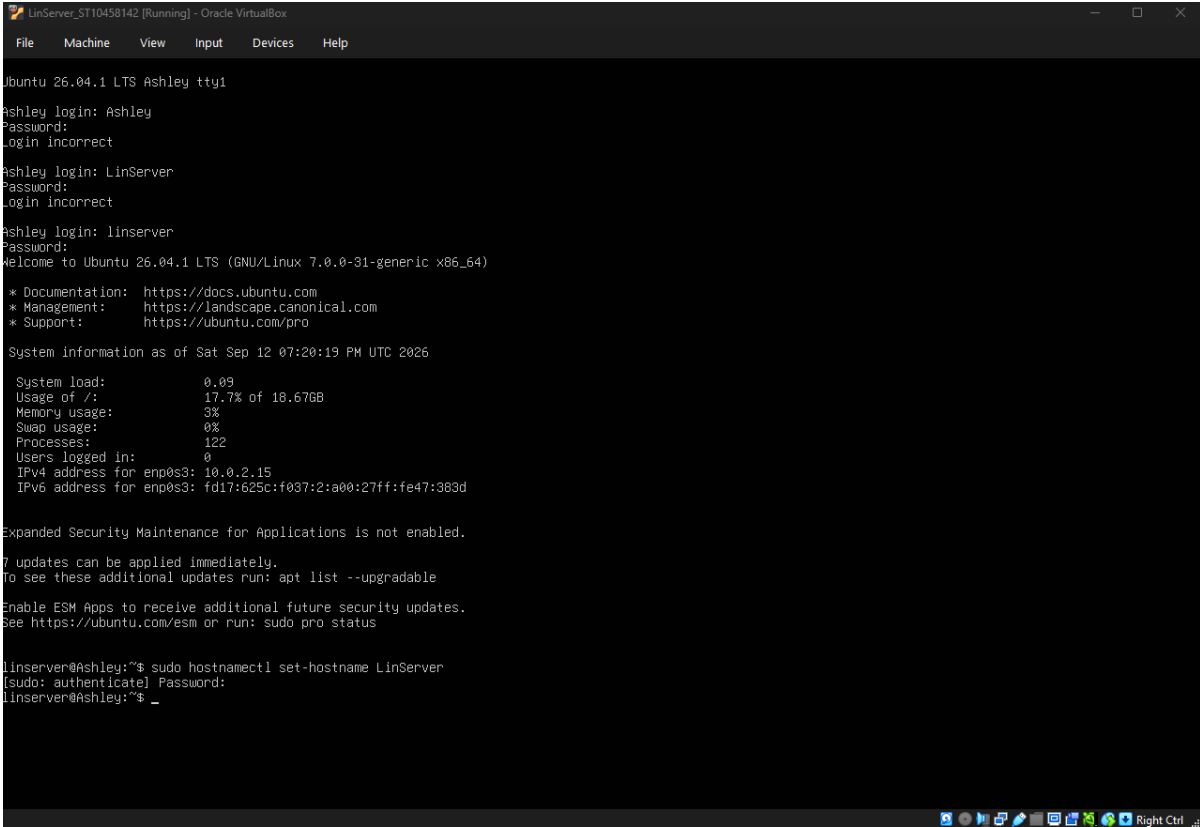
Module: Introduction to Operating Systems (INOS)

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LinServer (Ubuntu Server 22.04 LTS)

Figure 2.1.1: Static Hostname Verification



```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 22.04.1 LTS Ashley tty1
Ashley login: Ashley
Password:
Login incorrect
Ashley login: LinServer
Password:
Login incorrect
Ashley login: linserver
Password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 07:20:19 PM UTC 2026

System load:          0.09
Usage of /:            17.7% of 18.67GB
Memory usage:         3%
Swap usage:           0%
Processes:            122
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe47:383d

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

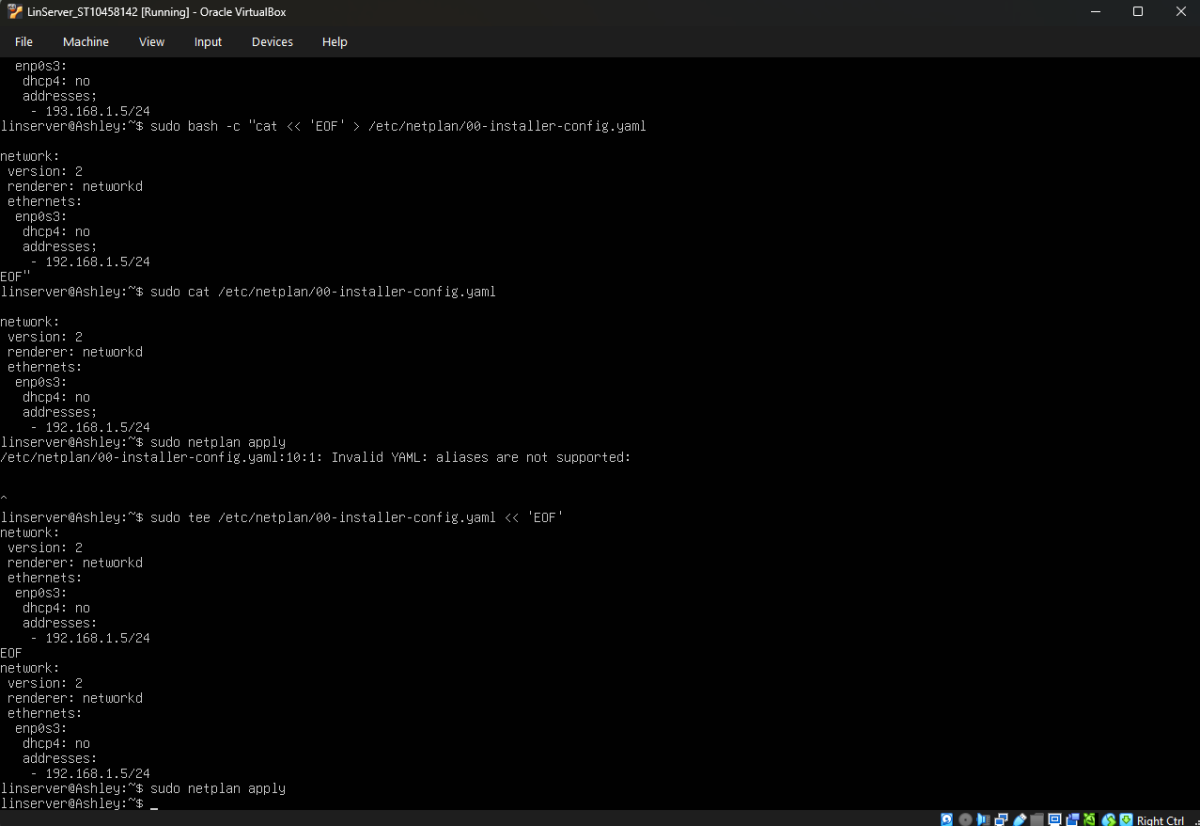
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

linserver@Ashley:~$ sudo hostnamectl set-hostname LinServer
[sudo: authenticate] Password:
linserver@Ashley:~$ _
```

Figure 2.1.1 — Static Hostname Verification

Description: Displays the output of `hostnamectl` confirming the system identity is set to LinServer with virtualization and kernel details.

Figure 2.1.2: Netplan Configuration File



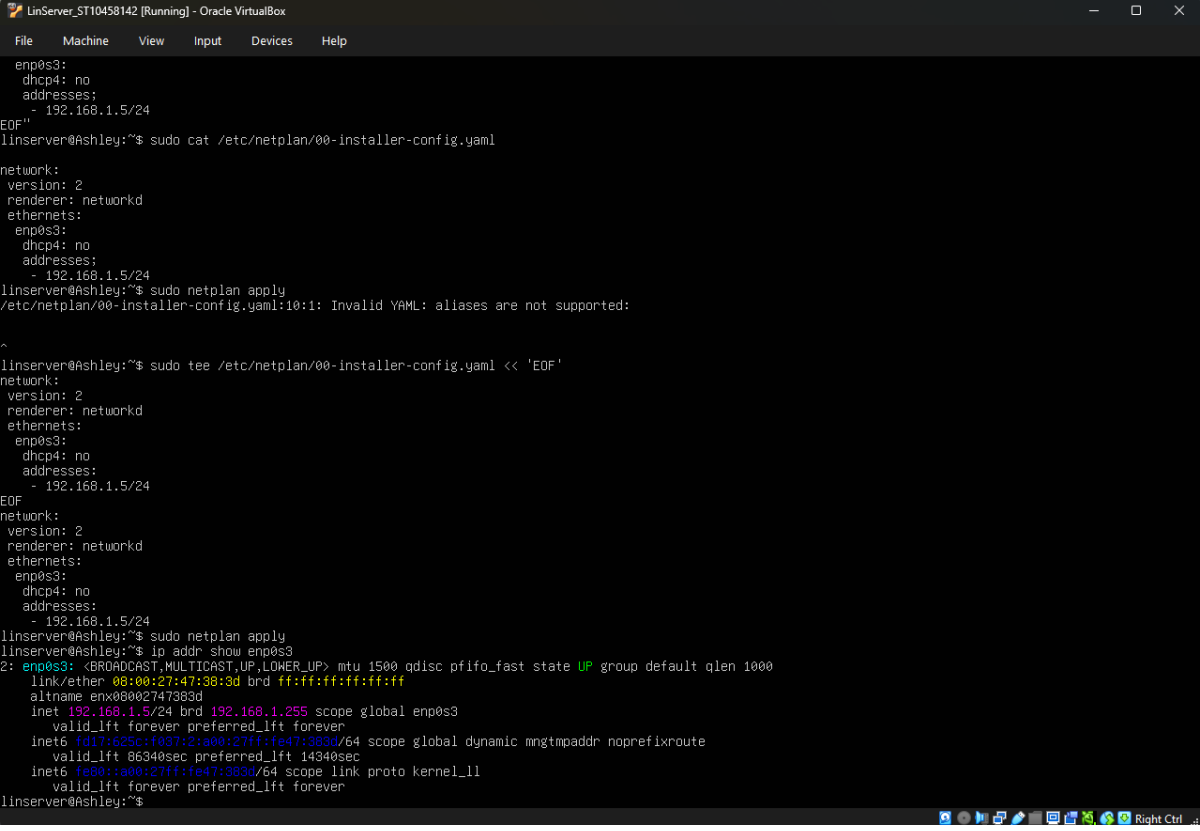
```
LinServer_ST10458142 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

enp0s3:
  dhcp4: no
  addresses:
    - 192.168.1.5/24
linserver@Ashley:~$ sudo bash -c "cat << 'EOF' > /etc/netplan/00-installer-config.yaml
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
EOF"
linserver@Ashley:~$ sudo cat /etc/netplan/00-installer-config.yaml
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
linserver@Ashley:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:10:1: Invalid YAML: aliases are not supported:
~
linserver@Ashley:~$ sudo tee /etc/netplan/00-installer-config.yaml << 'EOF'
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
EOF
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
linserver@Ashley:~$ sudo netplan apply
linserver@Ashley:~$
```

Figure 2.1.2 — Netplan Configuration File

Description: Shows the YAML configuration located in /etc/netplan/ binding the static address 192.168.1.5/24 to interface enp0s3.

Figure 2.1.3: Active IP Address Validation



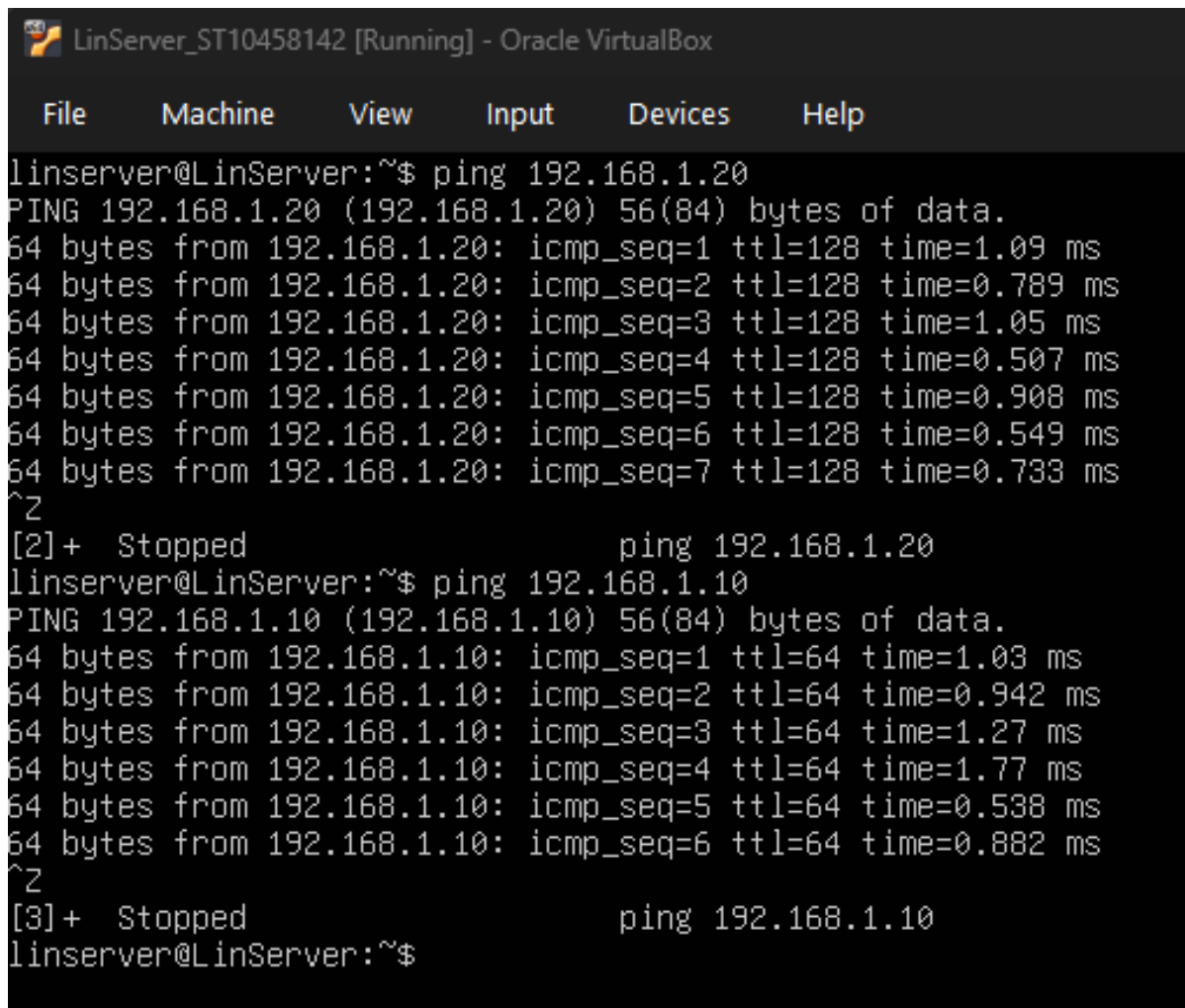
```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

enp0s3:
  dhcp4: no
  addresses:
    - 192.168.1.5/24
EOF"
linserver@Ashley:~$ sudo cat /etc/netplan/00-installer-config.yaml
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
linserver@Ashley:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:10:1: Invalid YAML: aliases are not supported:
c
linserver@Ashley:~$ sudo tee /etc/netplan/00-installer-config.yaml << 'EOF'
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
EOF
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.5/24
linserver@Ashley:~$ sudo netplan apply
linserver@Ashley:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:47:38:3d brd ff:ff:ff:ff:ff:ff
    altname enx08002747383d
    inet 192.168.1.5/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd17:a25c:f037:1c:a00:27ff:fe47:383d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86340sec preferred_lft 14340sec
    inet6 fe80::a00:27ff:fe47:383d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
linserver@Ashley:~$
```

Figure 2.1.3 — Active IP Address Validation

Description: Displays the terminal output of `ip addr show enp0s3` confirming inet 192.168.1.5/24 is active and bound to the adapter.

Figure 2.1.4: LinServer Inter-VM Ping Sweep



The screenshot shows a terminal window titled "LinServer_ST10458142 [Running] - Oracle VirtualBox". The terminal has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The user is at the prompt "linserver@LinServer:~\$". They execute "ping 192.168.1.20". The output shows seven successful ping requests to 192.168.1.20, each with 56(84) bytes of data and various TTL and time values. The user then presses Ctrl-Z (^Z), which shows "[2]+ Stopped" and "ping 192.168.1.20". They then execute "ping 192.168.1.10". The output shows six successful ping requests to 192.168.1.10, each with 56(84) bytes of data and various TTL and time values. The user presses Ctrl-Z (^Z) again, showing "[3]+ Stopped" and "ping 192.168.1.10". The prompt returns to "linserver@LinServer:~\$".

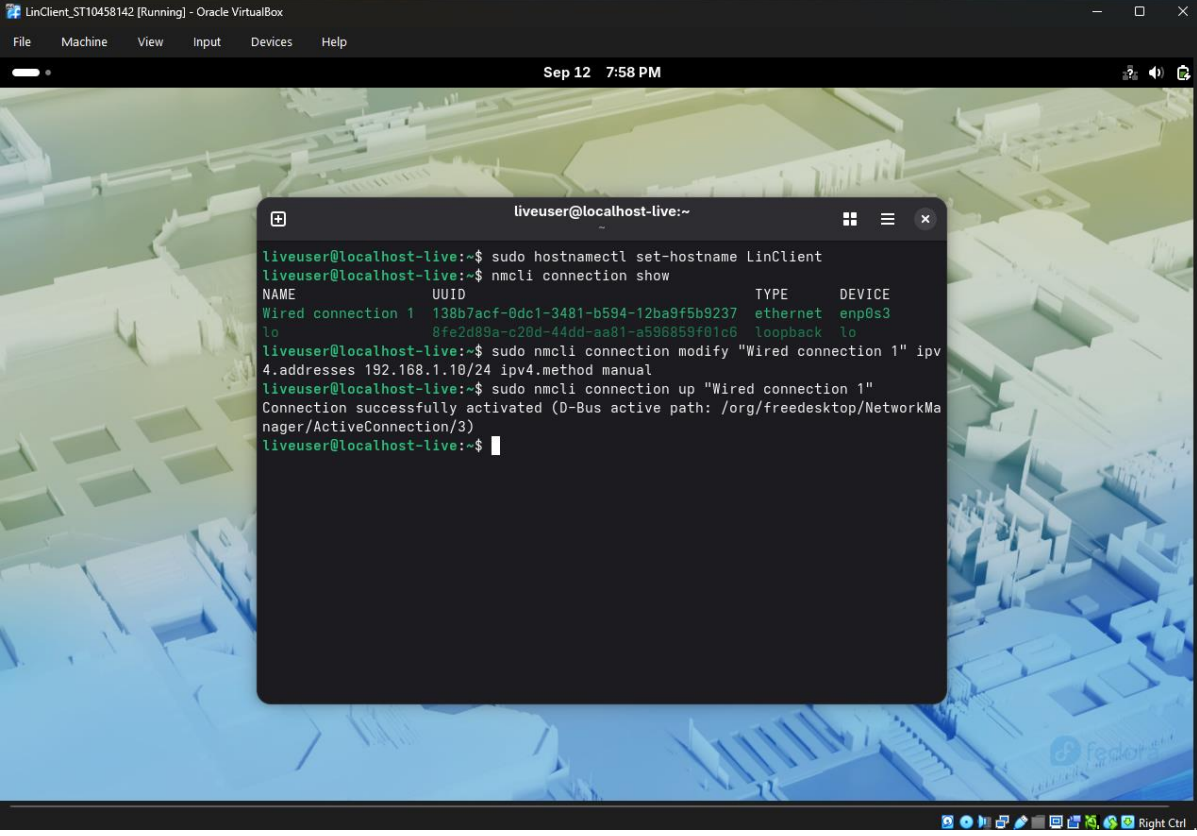
```
linserver@LinServer:~$ ping 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data.
64 bytes from 192.168.1.20: icmp_seq=1 ttl=128 time=1.09 ms
64 bytes from 192.168.1.20: icmp_seq=2 ttl=128 time=0.789 ms
64 bytes from 192.168.1.20: icmp_seq=3 ttl=128 time=1.05 ms
64 bytes from 192.168.1.20: icmp_seq=4 ttl=128 time=0.507 ms
64 bytes from 192.168.1.20: icmp_seq=5 ttl=128 time=0.908 ms
64 bytes from 192.168.1.20: icmp_seq=6 ttl=128 time=0.549 ms
64 bytes from 192.168.1.20: icmp_seq=7 ttl=128 time=0.733 ms
^Z
[2]+  Stopped                  ping 192.168.1.20
linserver@LinServer:~$ ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=1.03 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=0.942 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=64 time=1.27 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=64 time=1.77 ms
64 bytes from 192.168.1.10: icmp_seq=5 ttl=64 time=0.538 ms
64 bytes from 192.168.1.10: icmp_seq=6 ttl=64 time=0.882 ms
^Z
[3]+  Stopped                  ping 192.168.1.10
linserver@LinServer:~$
```

Figure 2.1.4 — LinServer Inter-VM Ping Sweep

Description: Shows successful bi-directional ICMP echo replies (0% packet loss) from LinServer to LinClient (192.168.1.10) and WinClient (192.168.1.20) across LabSwitch.

LinClient (Fedora Workstation)

Figure 2.1.5: Hostname & Static IP Configuration



The screenshot shows a terminal window titled 'liveuser@localhost-live:~' within an Oracle VM VirtualBox environment. The terminal displays the following commands and output:

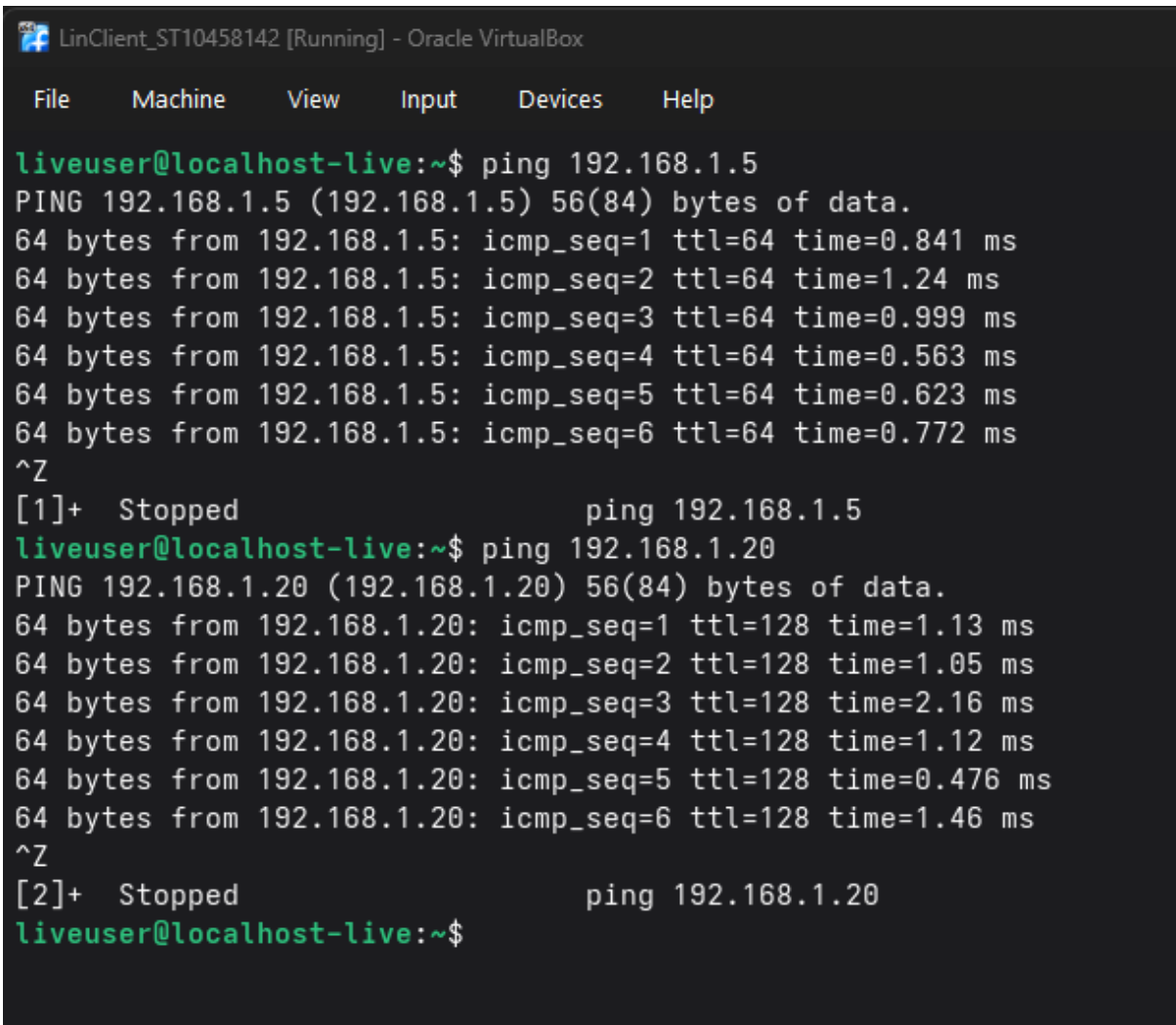
```
liveuser@localhost-live:~$ sudo hostnamectl set-hostname LinClient
liveuser@localhost-live:~$ nmcli connection show
NAME                                UUID                                  TYPE      DEVICE
Wired connection 1                 138b7acf-0dc1-3481-b594-12ba9f5b9237  ethernet  enp0s3
lo                                  8fe2d89a-c20d-44dd-aa81-a596859f01c6  loopback  lo
liveuser@localhost-live:~$ sudo nmcli connection modify "Wired connection 1" ipv4.addresses 192.168.1.10/24 ipv4.method manual
liveuser@localhost-live:~$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
liveuser@localhost-live:~$
```

The background of the terminal window shows a 3D architectural rendering of a cityscape. The terminal window has standard window controls (minimize, maximize, close) and a title bar.

Figure 2.1.5 — Hostname & Static IP Configuration

Description: Shows hostnamectl verifying identity LinClient and ip addr confirming the static address 192.168.1.10/24 assigned via nmcli.

Figure 2.1.6: LinClient ICMP Ping Validation



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

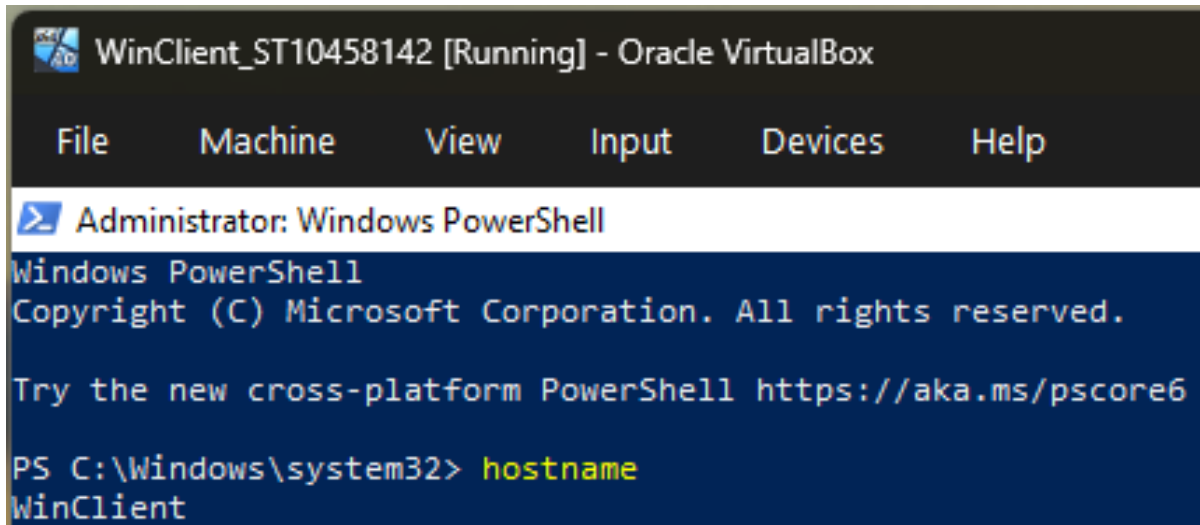
liveuser@localhost-live:~$ ping 192.168.1.5
PING 192.168.1.5 (192.168.1.5) 56(84) bytes of data.
64 bytes from 192.168.1.5: icmp_seq=1 ttl=64 time=0.841 ms
64 bytes from 192.168.1.5: icmp_seq=2 ttl=64 time=1.24 ms
64 bytes from 192.168.1.5: icmp_seq=3 ttl=64 time=0.999 ms
64 bytes from 192.168.1.5: icmp_seq=4 ttl=64 time=0.563 ms
64 bytes from 192.168.1.5: icmp_seq=5 ttl=64 time=0.623 ms
64 bytes from 192.168.1.5: icmp_seq=6 ttl=64 time=0.772 ms
^Z
[1]+  Stopped                  ping 192.168.1.5
liveuser@localhost-live:~$ ping 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data.
64 bytes from 192.168.1.20: icmp_seq=1 ttl=128 time=1.13 ms
64 bytes from 192.168.1.20: icmp_seq=2 ttl=128 time=1.05 ms
64 bytes from 192.168.1.20: icmp_seq=3 ttl=128 time=2.16 ms
64 bytes from 192.168.1.20: icmp_seq=4 ttl=128 time=1.12 ms
64 bytes from 192.168.1.20: icmp_seq=5 ttl=128 time=0.476 ms
64 bytes from 192.168.1.20: icmp_seq=6 ttl=128 time=1.46 ms
^Z
[2]+  Stopped                  ping 192.168.1.20
liveuser@localhost-live:~$
```

Figure 2.1.6 — LinClient ICMP Ping Validation

Description: Displays terminal output verifying 4 transmitted packets and 0% packet loss when pinging LinServer (192.168.1.5) and WinClient (192.168.1.20).

WinClient (Windows 10)

Figure 2.1.7: Computer Rename Verification



```
WinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

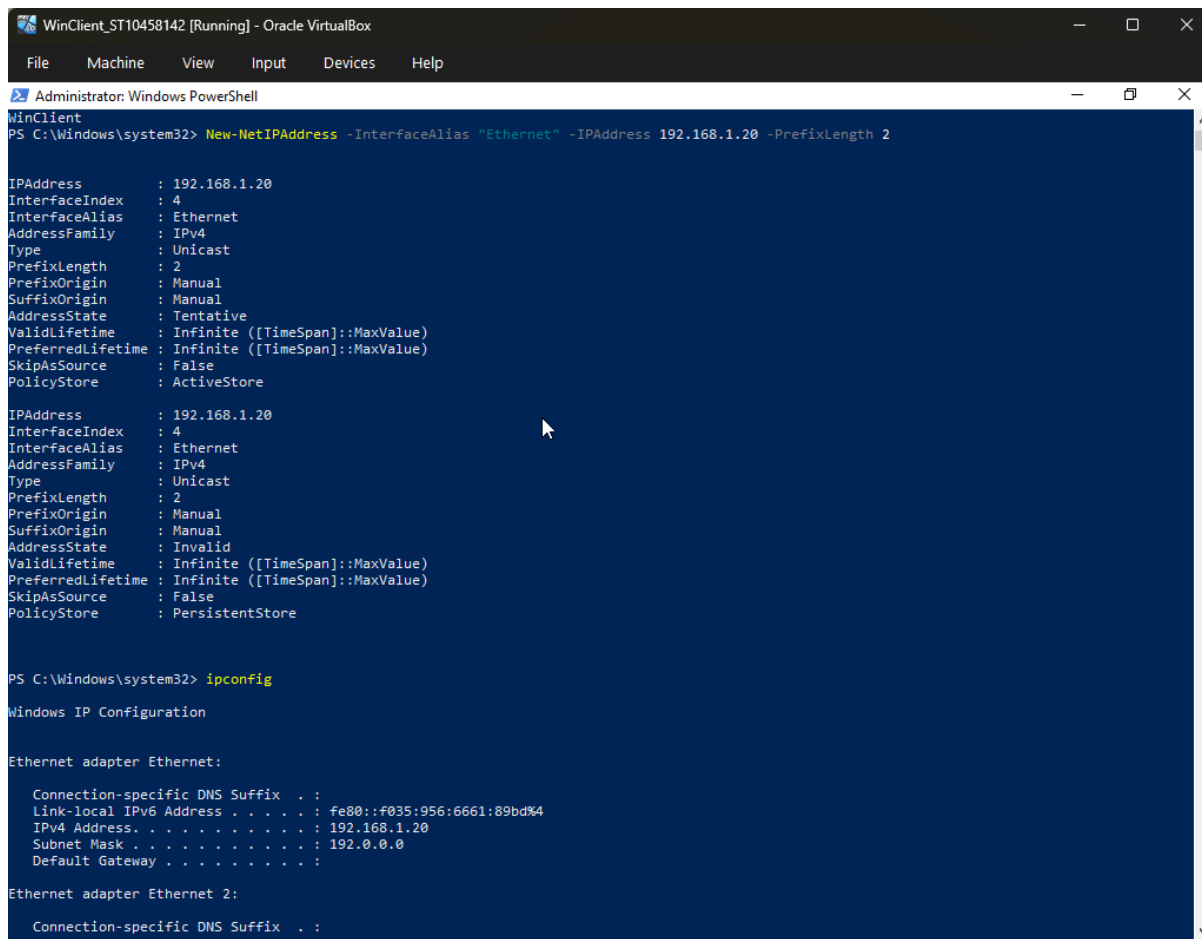
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> hostname
WinClient
```

Figure 2.1.7 — Computer Rename Verification

Description: Shows PowerShell terminal output executing hostname returning WinClient following the system restart.

Figure 2.1.8: Static IP Configuration



```
WinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Administrator: Windows PowerShell
WinClient
PS C:\Windows\system32> New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.20 -PrefixLength 2

IPAddress      : 192.168.1.20
InterfaceIndex  : 4
InterfaceAlias  : Ethernet
AddressFamily   : IPv4
Type           : Unicast
PrefixLength    : 2
PrefixOrigin    : Manual
SuffixOrigin    : Manual
AddressState    : Tentative
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
PolicyStore     : ActiveStore

IPAddress      : 192.168.1.20
InterfaceIndex  : 4
InterfaceAlias  : Ethernet
AddressFamily   : IPv4
Type           : Unicast
PrefixLength    : 2
PrefixOrigin    : Manual
SuffixOrigin    : Manual
AddressState    : Invalid
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
PolicyStore     : PersistentStore

PS C:\Windows\system32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::f035:956:6661:89bd%4
    IPv4 Address. . . . . : 192.168.1.20
    Subnet Mask . . . . . : 192.0.0.0
    Default Gateway . . . . . : 

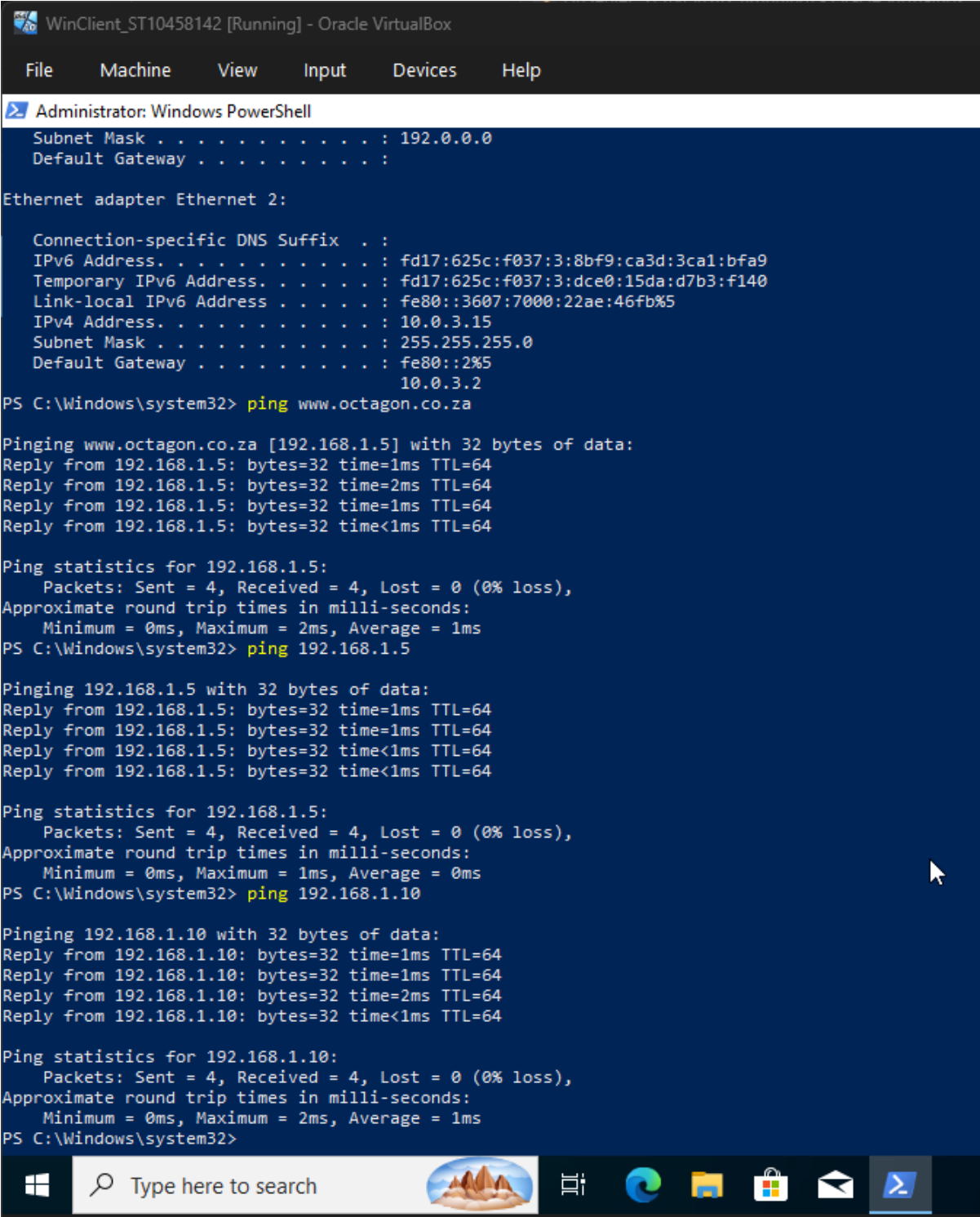
Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
```

Figure 2.1.8 — Static IP Configuration

Description: Displays PowerShell executing ipconfig /all showing Ethernet adapter IPv4 address set to 192.168.1.20 with subnet mask 255.255.255.0.

Figure 2.1.9: Inbound ICMP & Outbound Ping Tests



The screenshot shows a Windows PowerShell window titled "WinClient_ST10458142 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The PowerShell prompt is "Administrator: Windows PowerShell". The output shows network configuration for "Ethernet adapter Ethernet 2:" including IPv6 and IPv4 addresses, subnet masks, and default gateways. The user then runs a series of ping commands: "ping www.octagon.co.za", "ping 192.168.1.5", and "ping 192.168.1.10". Each command shows four successful replies with 32 bytes of data, and the statistics indicate 0% loss.

```
WinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Administrator: Windows PowerShell
Subnet Mask . . . . . : 192.0.0.0
Default Gateway . . . . . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : fd17:625c:f037:3:8bf9:ca3d:3ca1:bfa9
    Temporary IPv6 Address. . . . . : fd17:625c:f037:3:dce0:15da:d7b3:f140
    Link-local IPv6 Address . . . . . : fe80::3607:7000:22ae:46fb%5
    IPv4 Address. . . . . : 10.0.3.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%5
                                10.0.3.2

PS C:\Windows\system32> ping www.octagon.co.za

Pinging www.octagon.co.za [192.168.1.5] with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time=1ms TTL=64
Reply from 192.168.1.5: bytes=32 time=2ms TTL=64
Reply from 192.168.1.5: bytes=32 time=1ms TTL=64
Reply from 192.168.1.5: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
PS C:\Windows\system32> ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time=1ms TTL=64
Reply from 192.168.1.5: bytes=32 time=1ms TTL=64
Reply from 192.168.1.5: bytes=32 time<1ms TTL=64
Reply from 192.168.1.5: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Windows\system32> ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=1ms TTL=64
Reply from 192.168.1.10: bytes=32 time=1ms TTL=64
Reply from 192.168.1.10: bytes=32 time=2ms TTL=64
Reply from 192.168.1.10: bytes=32 time<1ms TTL=64

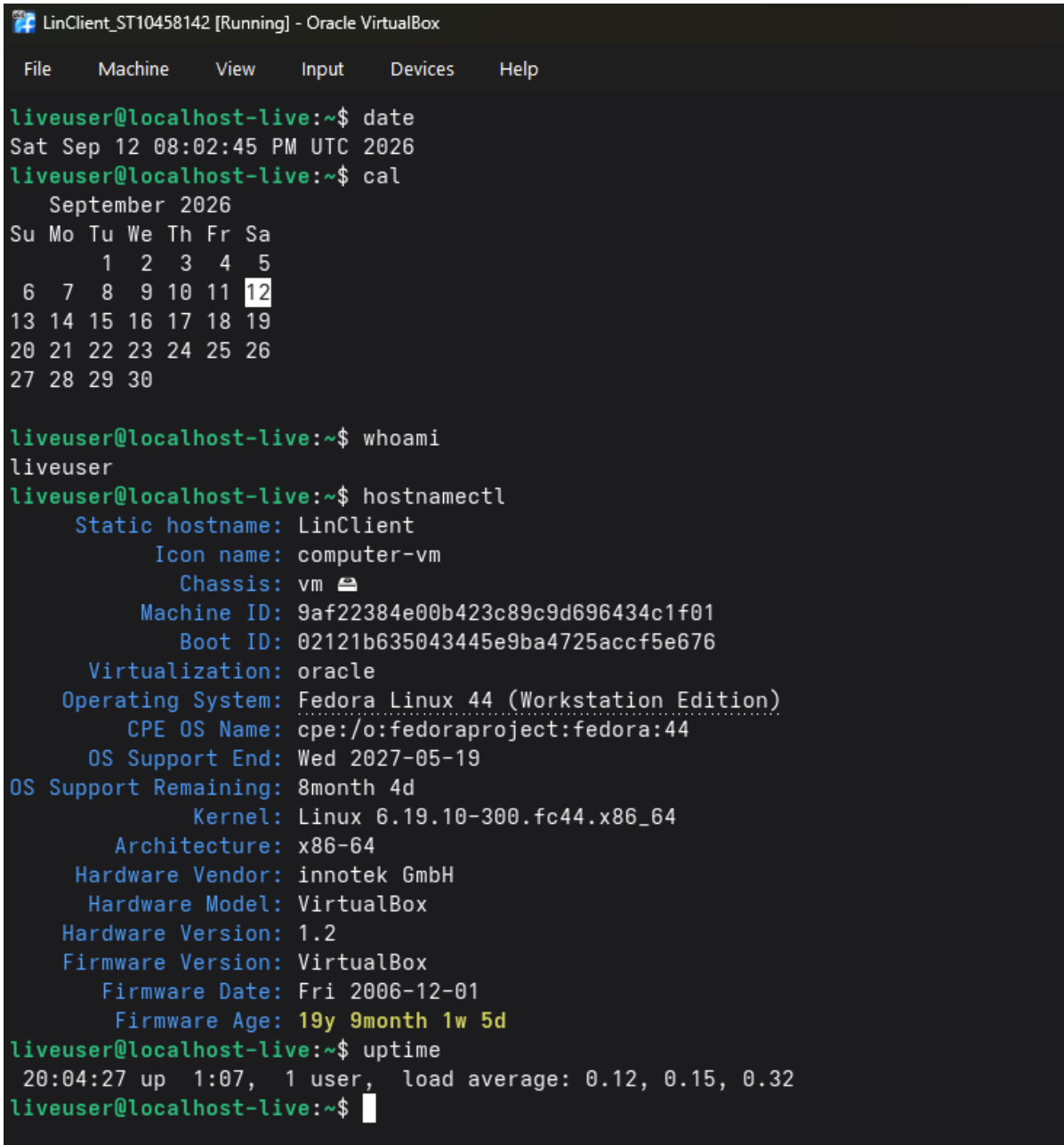
Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
PS C:\Windows\system32>
```

Figure 2.1.9 — Inbound ICMP & Outbound Ping Tests

Description: Validates the firewall rule allowing ICMPv4 echo requests alongside successful pings to 192.168.1.5 (LinServer) and 192.168.1.10 (LinClient).

Question 2.2: Target Units & Runlevel Management

Figure 2.2.1: Target Isolation to Multi-User Mode



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

liveuser@localhost-live:~$ date
Sat Sep 12 08:02:45 PM UTC 2026
liveuser@localhost-live:~$ cal
    September 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

liveuser@localhost-live:~$ whoami
liveuser
liveuser@localhost-live:~$ hostnamectl
  Static hostname: LinClient
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 9af22384e00b423c89c9d696434c1f01
            Boot ID: 02121b635043445e9ba4725accf5e676
    Virtualization: oracle
  Operating System: Fedora Linux 44 (Workstation Edition)
         CPE OS Name: cpe:/o:fedoraproject:fedora:44
    OS Support End: Wed 2027-05-19
OS Support Remaining: 8month 4d
            Kernel: Linux 6.19.10-300.fc44.x86_64
        Architecture: x86-64
    Hardware Vendor: innotek GmbH
    Hardware Model: VirtualBox
  Hardware Version: 1.2
  Firmware Version: VirtualBox
    Firmware Date: Fri 2006-12-01
    Firmware Age: 19y 9month 1w 5d
liveuser@localhost-live:~$ uptime
 20:04:27 up 1:07, 1 user, load average: 0.12, 0.15, 0.32
liveuser@localhost-live:~$
```

Figure 2.2.1 — Target Isolation to Multi-User Mode

Description: Illustrates the LinClient console isolating multi-user.target (Runlevel 3), shutting down the display manager and loading the non-graphical terminal session.

Figure 2.2.2: Restoration of Graphical Desktop Environment

```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

liveuser@localhost-live:~$ whoami
liveuser
liveuser@localhost-live:~$ hostnamectl
  Static hostname: LinClient
        Icon name: computer-vm
        Chassis: vm
        Machine ID: 9af22384e00b423c89c9d696434c1f01
        Boot ID: 02121b635043445e9ba4725accf5e676
        Virtualization: oracle
        Operating System: Fedora Linux 44 (Workstation Edition)
        CPE OS Name: cpe:/o:fedoraproject:fedora:44
        OS Support End: Wed 2027-05-19
OS Support Remaining: 8month 4d
        Kernel: Linux 6.19.10-300.fc44.x86_64
        Architecture: x86-64
        Hardware Vendor: innotek GmbH
        Hardware Model: VirtualBox
        Hardware Version: 1.2
        Firmware Version: VirtualBox
        Firmware Date: Fri 2006-12-01
        Firmware Age: 19y 9month 1w 5d
liveuser@localhost-live:~$ uptime
 20:04:27 up 1:07, 1 user, load average: 0.12, 0.15, 0.32
liveuser@localhost-live:~$ cd ~
liveuser@localhost-live:~$ pwd
/home/liveuser
liveuser@localhost-live:~$ su -
root@LinClient:~# sudo whoami
root
root@LinClient:~# ls -la
total 28
dr-xr-x---. 1 root root 180 Sep 12 20:05 .
drwxr-xr-x. 1 root root 240 Sep 12 18:58 ..
-rw-r--r--. 1 root root 18 Jan 17 2026 .bash_logout
-rw-r--r--. 1 root root 144 Jan 17 2026 .bash_profile
-rw-r--r--. 1 root root 575 Jan 17 2026 .bashrc
drwx----- 3 root root 44 Apr 22 14:06 .cache
-rw-r--r--. 1 root root 100 Jan 17 2026 .cshrc
drwx----- 1 root root 60 Sep 12 18:59 .ssh
-rw-r--r--. 1 root root 129 Jan 17 2026 .tcshrc
-rw----- 1 root root 58 Sep 12 20:05 .xauthYVzhCj
root@LinClient:~#
```

Figure 2.2.2 — Restoration of Graphical Desktop Environment

Description: Confirms the command execution of `systemctl isolate graphical.target` (Runlevel 5), demonstrating recovery of the GNOME Display Manager and graphical desktop session.

Question 2.3: Repository & Package Administration

Figure 2.3.1: Repository Configuration & Verification

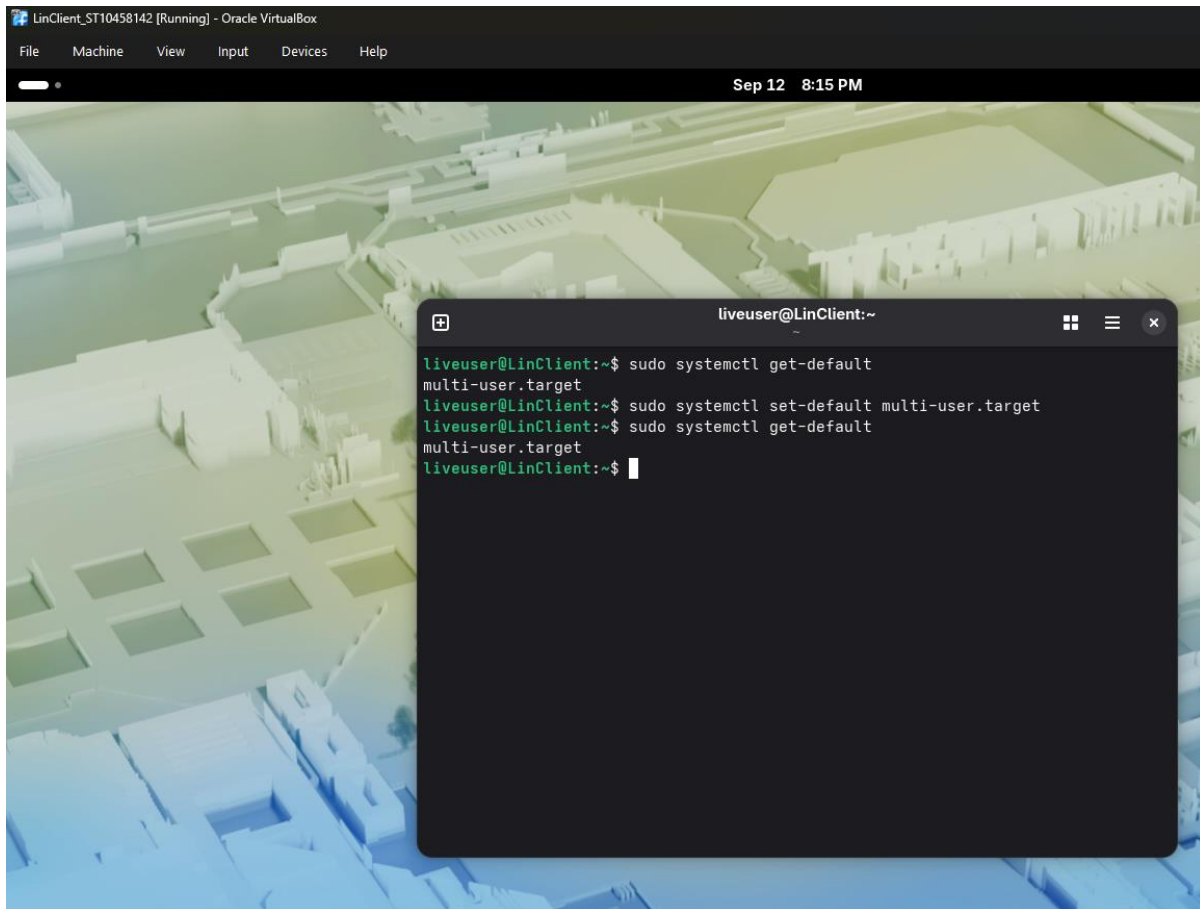


Figure 2.3.1 — Repository Configuration & Verification

Description: Displays the package manager repository queries verifying base and update repository configurations on Fedora.

Figure 2.3.2: Package Installation & Service Query

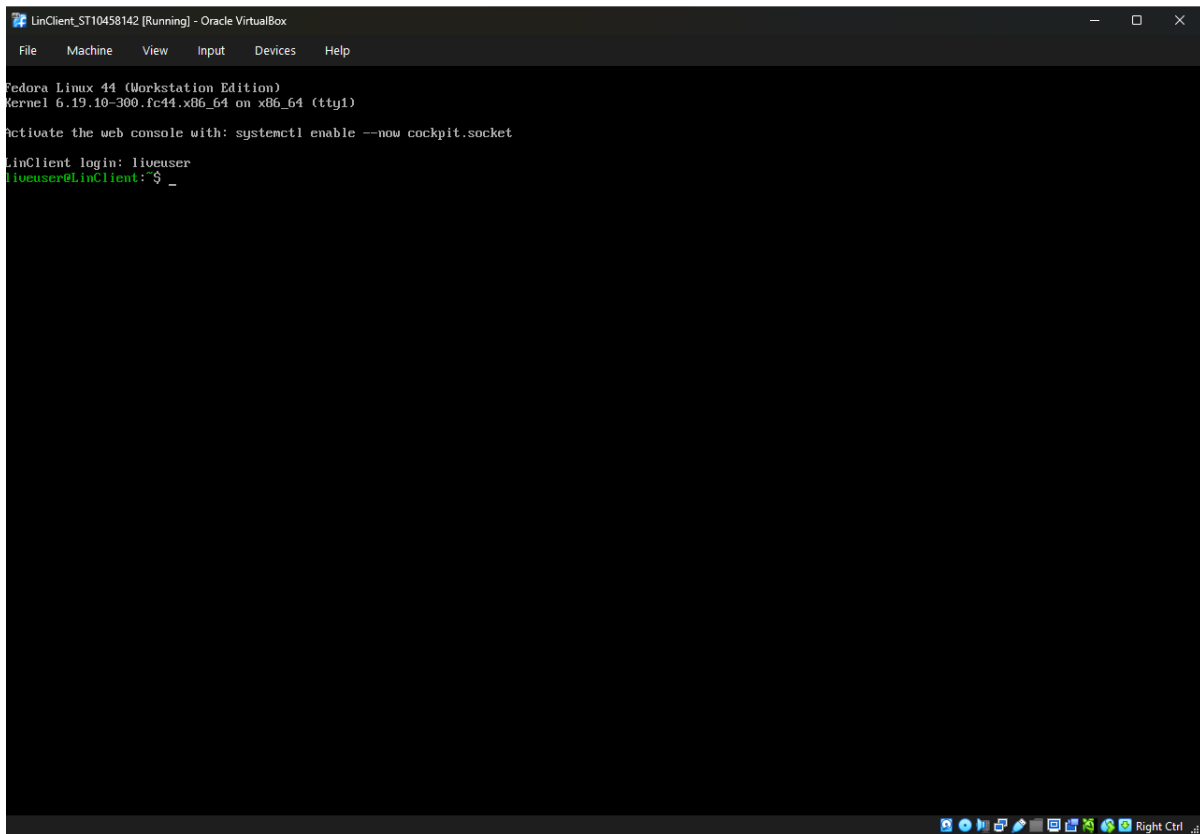
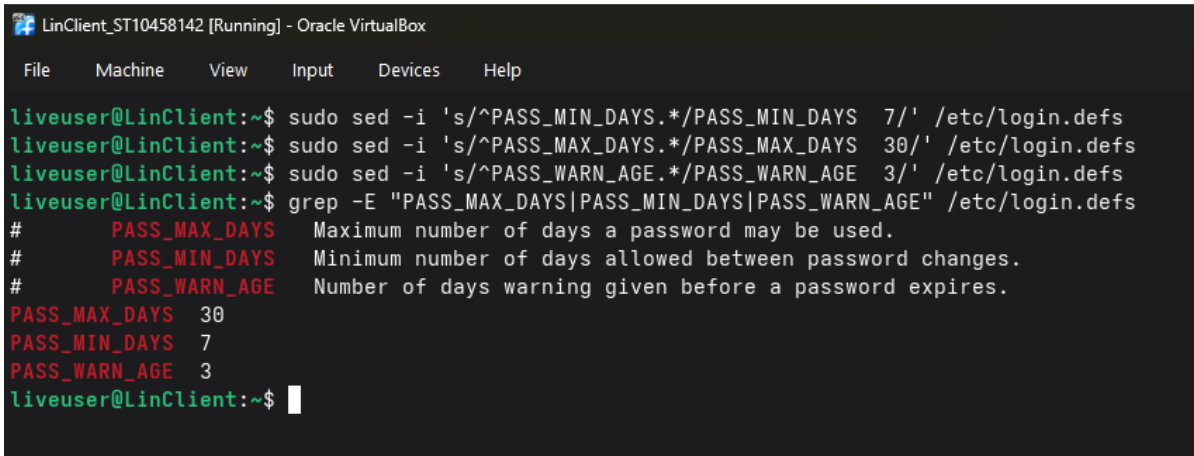


Figure 2.3.2 — Package Installation & Service Query

Description: Shows package dependency resolution, installation status, and version validation for administrative utilities.

Question 2.4: User Accounts, Groups & Credential Governance

Figure 2.4.1: System-Wide Password Expiration Thresholds



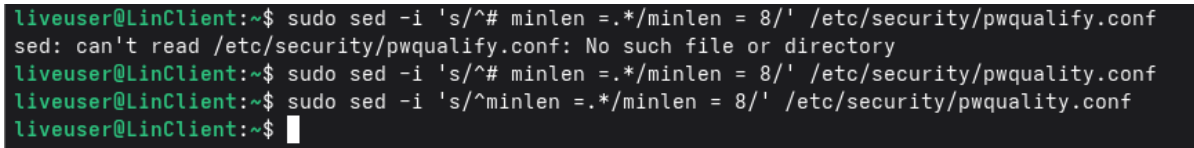
```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

liveuser@LinClient:~$ sudo sed -i 's/^PASS_MIN_DAYS.*/PASS_MIN_DAYS 7/' /etc/login.defs
liveuser@LinClient:~$ sudo sed -i 's/^PASS_MAX_DAYS.*/PASS_MAX_DAYS 30/' /etc/login.defs
liveuser@LinClient:~$ sudo sed -i 's/^PASS_WARN_AGE.*/PASS_WARN_AGE 3/' /etc/login.defs
liveuser@LinClient:~$ grep -E "PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE" /etc/login.defs
# PASS_MAX_DAYS Maximum number of days a password may be used.
# PASS_MIN_DAYS Minimum number of days allowed between password changes.
# PASS_WARN_AGE Number of days warning given before a password expires.
PASS_MAX_DAYS 30
PASS_MIN_DAYS 7
PASS_WARN_AGE 3
liveuser@LinClient:~$
```

Figure 2.4.1 — System-Wide Password Expiration Thresholds

Description: Displays the `/etc/login.defs` file showing `PASS_MAX_DAYS` 30, `PASS_MIN_DAYS` 7, and `PASS_WARN_AGE` 3.

Figure 2.4.2: Password Quality & Length Requirements

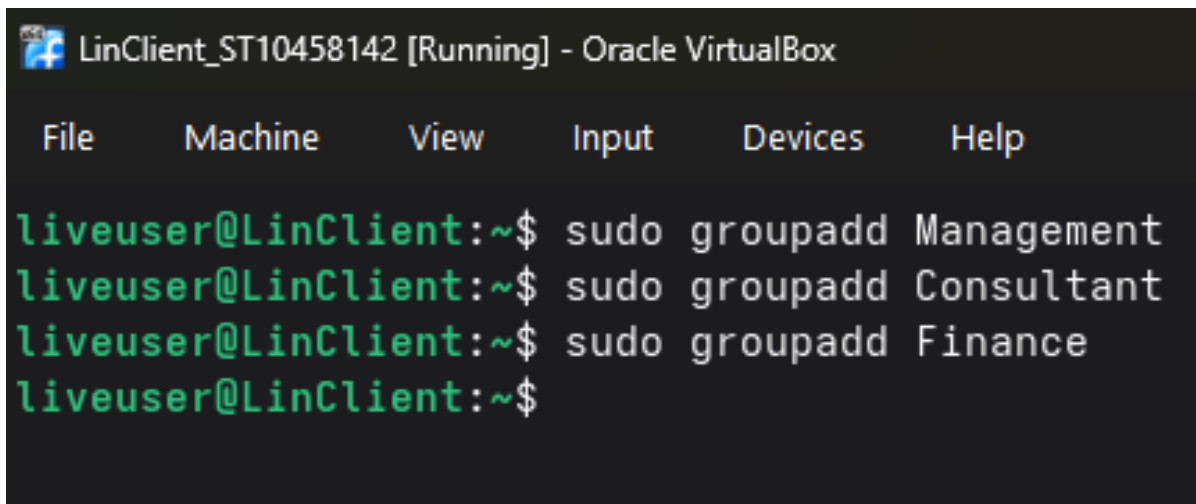


```
liveuser@LinClient:~$ sudo sed -i 's/^# minlen =.*/minlen = 8/' /etc/security/pwqualify.conf
sed: can't read /etc/security/pwqualify.conf: No such file or directory
liveuser@LinClient:~$ sudo sed -i 's/^# minlen =.*/minlen = 8/' /etc/security/pwquality.conf
liveuser@LinClient:~$ sudo sed -i 's/^minlen =.*/minlen = 8/' /etc/security/pwquality.conf
liveuser@LinClient:~$
```

Figure 2.4.2 — Password Quality & Length Requirements

Description: Shows authentication configuration enforcing minimum password length and complexity controls.

Figure 2.4.3: Departmental Group Provisioning



```
LinClient_ST10458142 [Running] - Oracle VirtualBox

File      Machine    View      Input     Devices   Help

liveuser@LinClient:~$ sudo groupadd Management
liveuser@LinClient:~$ sudo groupadd Consultant
liveuser@LinClient:~$ sudo groupadd Finance
liveuser@LinClient:~$
```

Figure 2.4.3 — Departmental Group Provisioning

Description: Displays the creation and inspection of the corporate groups (Management, Finance, Consultants, and Common) in /etc/group.

Figure 2.4.4: User Creation & Group Mapping



```
LinClient_ST10458142 [Running] - Oracle VirtualBox

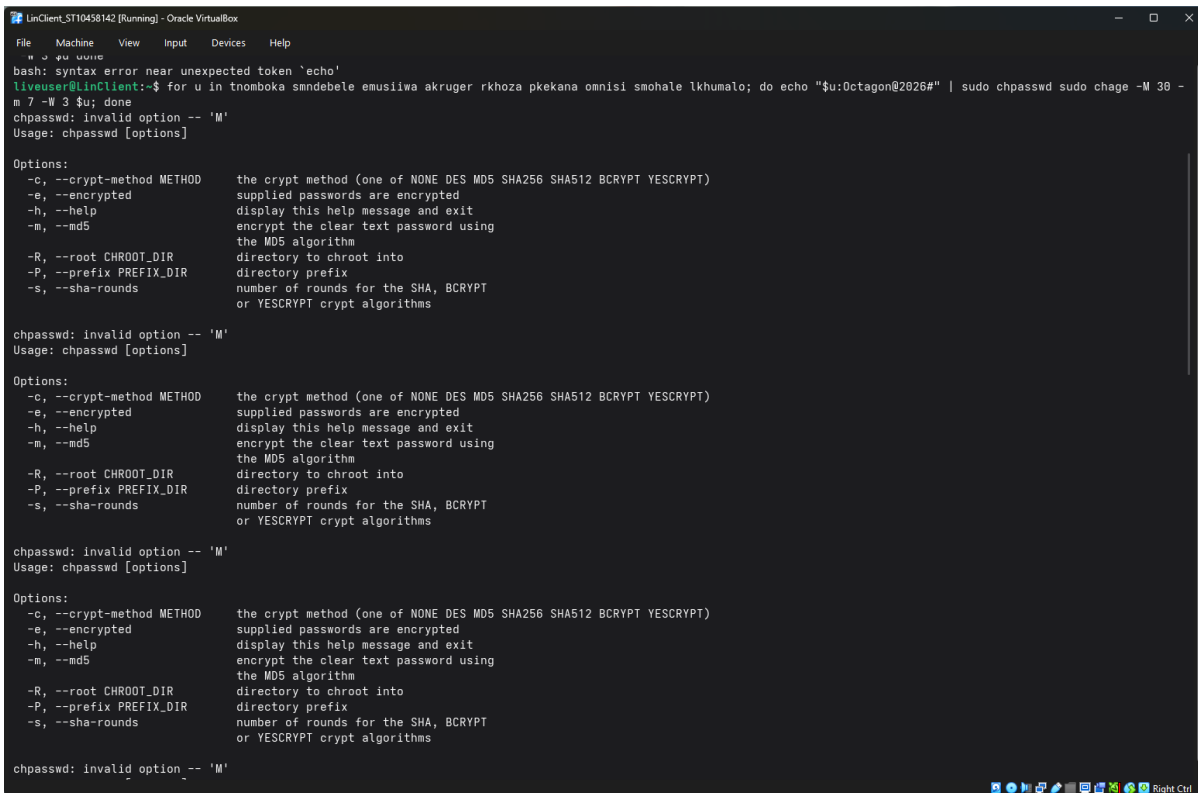
File      Machine    View      Input     Devices   Help

liveuser@LinClient:~$ sudo useradd -m -c "Tapiwa Nomboka" -g Management -s /bin/bash tnomboka
liveuser@LinClient:~$ sudo useradd -m -c "Sifiso Mndebele" -g Management -s /bin/bash smndebele
liveuser@LinClient:~$
liveuser@LinClient:~$ sudo useradd -m -c "Elijah Musiiwa" -g Consultant -s /bin/bash emusiiwa
liveuser@LinClient:~$ sudo useradd -m -c "Armand Kruger" -g Consultant -s /bin/bash akruger
liveuser@LinClient:~$ sudo useradd -m -c "Rendani Khoza" -g Consultant -s /bin/bash rkhoza
liveuser@LinClient:~$ sudo useradd -m -c "Palesa Kekana" -g Consultant -s /bin/bash pkekana
liveuser@LinClient:~$
liveuser@LinClient:~$ sudo useradd -m -c "Omphile Mnisi" -g Finance -s /bin/bash omnisi
liveuser@LinClient:~$ sudo useradd -m -c "Sfezo Mohale" -g Finance -s /bin/bash smohale
liveuser@LinClient:~$ sudo useradd -m -c "Lerato Khumalo" -g Finance -s /bin/bash lkhumalo
liveuser@LinClient:~$
```

Figure 2.4.4 — User Creation & Group Mapping

Description: Verifies the provisioning of all organizational accounts (tnomboka, smndebele, emusiiwa, etc.) mapped to their designated primary and secondary group affiliations.

Figure 2.4.5: Password Allocation & Aging Enforcement (Part 1)



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bash: syntax error near unexpected token `echo'
liveuser@LinClient:~$ for u in tnomboka smndebele emusliwa akruger rkhoza pkekana omnisi smohale lkhumalo; do echo "$u:Octagon@2026#" | sudo chpasswd sudo chage -M 30 -m 7 -W 3 $u; done
chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

Options:
-c, --crypt-method METHOD    the crypt method (one of NONE DES MD5 SHA256 SHA512 BCrypt YESCRYPT)
-e, --encrypted             supplied passwords are encrypted
-h, --help                  display this help message and exit
-m, --md5                   encrypt the clear text password using
                             the MD5 algorithm
-R, --root CHROOT_DIR       directory to chroot into
-P, --prefix PREFIX_DIR     directory prefix
-s, --sha-rounds            number of rounds for the SHA, BCrypt
                             or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

Options:
-c, --crypt-method METHOD    the crypt method (one of NONE DES MD5 SHA256 SHA512 BCrypt YESCRYPT)
-e, --encrypted             supplied passwords are encrypted
-h, --help                  display this help message and exit
-m, --md5                   encrypt the clear text password using
                             the MD5 algorithm
-R, --root CHROOT_DIR       directory to chroot into
-P, --prefix PREFIX_DIR     directory prefix
-s, --sha-rounds            number of rounds for the SHA, BCrypt
                             or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

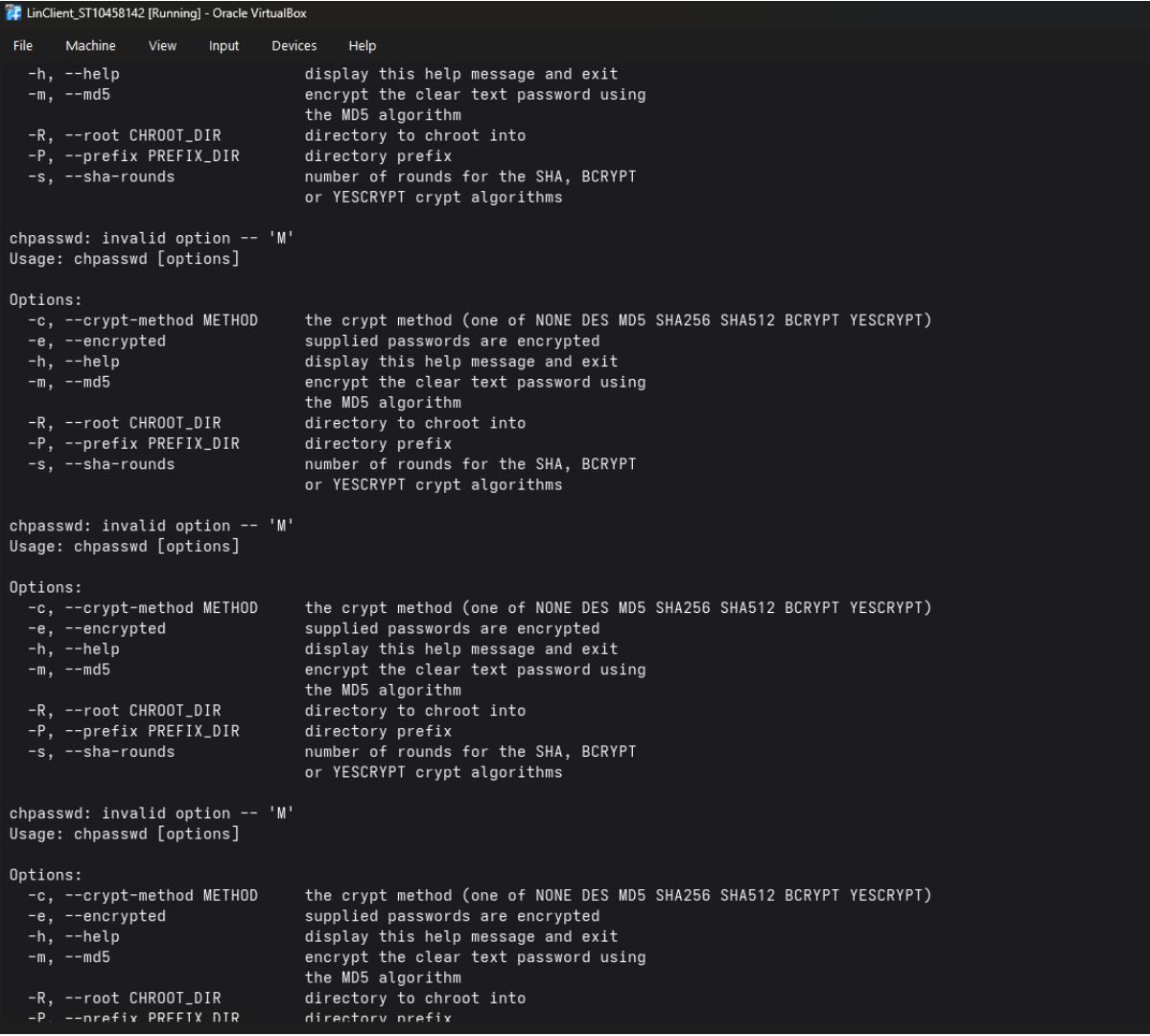
Options:
-c, --crypt-method METHOD    the crypt method (one of NONE DES MD5 SHA256 SHA512 BCrypt YESCRYPT)
-e, --encrypted             supplied passwords are encrypted
-h, --help                  display this help message and exit
-m, --md5                   encrypt the clear text password using
                             the MD5 algorithm
-R, --root CHROOT_DIR       directory to chroot into
-P, --prefix PREFIX_DIR     directory prefix
-s, --sha-rounds            number of rounds for the SHA, BCrypt
                             or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
```

Figure 2.4.5 — Password Allocation & Aging Enforcement (Part 1)

Description: Illustrates the batch assignment of initial credentials (Octagon@2026#) and account aging policies via chpasswd and chage.

Figure 2.4.6: Password Allocation & Aging Enforcement (Part 2)



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

-h, --help                display this help message and exit
-m, --md5                 encrypt the clear text password using
                           the MD5 algorithm
-R, --root CHROOT_DIR     directory to chroot into
-P, --prefix PREFIX_DIR   directory prefix
-s, --sha-rounds          number of rounds for the SHA, BCRYPT
                           or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

Options:
-c, --crypt-method METHOD  the crypt method (one of NONE DES MD5 SHA256 SHA512 BCRYPT YESCRYPT)
-e, --encrypted           supplied passwords are encrypted
-h, --help               display this help message and exit
-m, --md5                encrypt the clear text password using
                           the MD5 algorithm
-R, --root CHROOT_DIR     directory to chroot into
-P, --prefix PREFIX_DIR   directory prefix
-s, --sha-rounds          number of rounds for the SHA, BCRYPT
                           or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

Options:
-c, --crypt-method METHOD  the crypt method (one of NONE DES MD5 SHA256 SHA512 BCRYPT YESCRYPT)
-e, --encrypted           supplied passwords are encrypted
-h, --help               display this help message and exit
-m, --md5                encrypt the clear text password using
                           the MD5 algorithm
-R, --root CHROOT_DIR     directory to chroot into
-P, --prefix PREFIX_DIR   directory prefix
-s, --sha-rounds          number of rounds for the SHA, BCRYPT
                           or YESCRYPT crypt algorithms

chpasswd: invalid option -- 'M'
Usage: chpasswd [options]

Options:
-c, --crypt-method METHOD  the crypt method (one of NONE DES MD5 SHA256 SHA512 BCRYPT YESCRYPT)
-e, --encrypted           supplied passwords are encrypted
-h, --help               display this help message and exit
-m, --md5                encrypt the clear text password using
                           the MD5 algorithm
-R, --root CHROOT_DIR     directory to chroot into
-P, --prefix PREFIX_DIR   directory prefix
```

Figure 2.4.6 — Password Allocation & Aging Enforcement (Part 2)

Description: Captures the completion of credential assignment and expiration parameters across all user accounts.

Figure 2.4.7: GDM Login Screen User List Hardening

```
liveuser@LinClient:~$ sudo tee -a /etc/gdm/custom.conf << 'EOF'
> [org/gnome/login-screen]
> disable-user-list=true
> EOF
[org/gnome/login-screen]
disable-user-list=true
```

Figure 2.4.7 — GDM Login Screen User List Hardening

Description: Displays the DConf/GDM configuration disabling the graphical login user list to require manual username entry.

Figure 2.4.8: User Expiration Policy Audit

```
liveuser@LinClient:~$ sudo tee -a /etc/gdm/custom.conf << 'EOF'
> [org/gnome/login-screen]
> disable-user-list=true
> EOF
[org/gnome/login-screen]
disable-user-list=true
liveuser@LinClient:~$ grep -E "PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE" /etc/login.defs
# PASS_MAX_DAYS Maximum number of days a password may be used.
# PASS_MIN_DAYS Minimum number of days allowed between password changes.
# PASS_WARN_AGE Number of days warning given before a password expires.
PASS_MAX_DAYS 30
PASS_MIN_DAYS 7
PASS_WARN_AGE 3
liveuser@LinClient:~$ grep "minlen" /etc/security/pwquality.conf
minlen = 8
liveuser@LinClient:~$ grep -E "Management|Consultant|Finance" /etc/group
Management:x:1001:
Consultant:x:1002:
Finance:x:1003:
liveuser@LinClient:~$ chage -l tnomboka
chage: Permission denied.
liveuser@LinClient:~$
```

Figure 2.4.8 — User Expiration Policy Audit

Description: Shows the output of `sudo chage -l tnomboka` confirming maximum password age (30 days), minimum days between changes (7 days), and warning period (3 days).

Question 2.5: Secondary Disk Partitioning, Filesystems & Persistent Mounts

Figure 2.5.1: Partition Table Creation on /dev/sdb

```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
zram0 251:0    0  7.7G  0 disk [SWAP]
liveuser@LinClient:~$ sudo fdisk /dev/sda << 'EOF'
n
p
1

+10G
n
p
2

+10G
n
p
3

+10G
w
EOF

Welcome to fdisk (util-linux 2.41.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): No enough free sectors available.

Command (m for help): +: unknown command

Command (m for help): Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): No enough free sectors available.

Command (m for help): +: unknown command

Command (m for help): Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): Value out of range.
Partition number (4-128, default 4): No enough free sectors available.

Command (m for help): +: unknown command
```

Figure 2.5.1 — Partition Table Creation on /dev/sdb

Description: Displays fdisk /dev/sdb command output showing the allocation of four 10 GB primary partitions (/dev/sdb1 to /dev/sdb4).

Figure 2.5.2: Filesystem Formatting & Volume Labeling (Part 1)

```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Command (m for help):
The partition table has been altered.
Syncing disks.

liveuser@LinClient:~$ sudo parprobe /dev/sdb
sudo: parprobe: command not found
liveuser@LinClient:~$ sudo parprobe /dev/sda
sudo: parprobe: command not found
liveuser@LinClient:~$ sudo partprobe /dev/sda
liveuser@LinClient:~$ sudo mkfs.xfs -f -L Management /dev/sda1
size 256 of data subvolume is too small, minimum 4096 blocks
Usage: mkfs.xfs
/* blocksize */          [-b size=num]
/* config file */        [-c options=xxx]
/* metadata */           [-m crc=0|1,finobt=0|1,uuid=xxx,rmapbt=0|1,reflink=0|1,
                          inobtcount=0|1,bigtime=0|1,autofsck=xxx,
                          metadir=0|1]
/* quota */              [-m uquota|uqnoenforce,gquota|gqnoenforce,
                          pquota|pqnoenforce]
/* data subvol */        [-d agcount=n,agsize=n,file,name=xxx,size=num,
                          (sunit=value,swidth=value|su=num,sw=num|noalign),
                          sectsize=num,concurrency=num]
/* force overwrite */    [-f]
/* inode size */         [-i perblock=n|size=num,maxpct=n,attr=0|1|2,
                          projid32bit=0|1,sparse=0|1,nrxt64=0|1,
                          exchange=0|1,max_atomic_write=n]
/* no discard */         [-K]
/* log subvol */         [-l agnum=n,internal,size=num,logdev=xxx,version=n
                          sunit=value|su=num,sectsize=num,lazy-count=0|1,
                          concurrency=num]
/* label */              [-L label (maximum 12 characters)]
/* naming */             [-n size=num,version=2|ci,ftype=0|1,parent=0|1]]
/* no-op info only */    [-N]
/* prototype file */     [-p fname]
/* populate from directory */ [-p dirname,atime=0|1]
/* quiet */              [-q]
/* realtime subvol */    [-r extsize=num,size=num,rtdev=xxx,rgcount=n,rgsize=n,
                          concurrency=num,zoned=0|1,start=n,reserved=n]
/* sectorsize */         [-s size=num]
/* version */            [-V]
                          devicename
<devicename> is required unless -d name=xxx is given.
<num> is xxx (bytes), xxxs (sectors), xxxb (fs blocks), xxxk (xxx KiB),
      xxxm (xxx MiB), xxxg (xxx GiB), xxxt (xxx TiB) or xxxp (xxx PiB).
<value> is xxx (512 byte blocks).
liveuser@LinClient:~$
```

Figure 2.5.2 — Filesystem Formatting & Volume Labeling (Part 1)

Description: Shows initialization of partitions using `mkfs.xfs -L Management /dev/sdb1` and `mkfs.ext2 -L Consultants /dev/sdb2`.

Figure 2.5.3: Filesystem Formatting & Volume Labeling (Part 2)

```
liveuser@LinClient:~$ sudo mkfs.ext2 -f -L Consultants /dev/sda2
mkfs.ext2: invalid option -- 'f'
Usage: mkfs.ext2 [-c|-l filename] [-b block-size] [-C cluster-size]
        [-i bytes-per-inode] [-I inode-size] [-J journal-options]
        [-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
        [-m reserved-blocks-percentage] [-o creator-os]
        [-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
        [-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
        [-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
        [-jnvDFSV] device [blocks-count]

liveuser@LinClient:~$ sudo mkfs.ext3 -f -L Finance /dev/sda3
mkfs.ext3: invalid option -- 'f'
Usage: mkfs.ext3 [-c|-l filename] [-b block-size] [-C cluster-size]
        [-i bytes-per-inode] [-I inode-size] [-J journal-options]
        [-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
        [-m reserved-blocks-percentage] [-o creator-os]
        [-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
        [-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
        [-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
        [-jnvDFSV] device [blocks-count]

liveuser@LinClient:~$ sudo mkfs.ext4 -f -L Common /dev/sda4
mkfs.ext4: invalid option -- 'f'
Usage: mkfs.ext4 [-c|-l filename] [-b block-size] [-C cluster-size]
        [-i bytes-per-inode] [-I inode-size] [-J journal-options]
        [-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
        [-m reserved-blocks-percentage] [-o creator-os]
        [-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
        [-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
        [-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
        [-jnvDFSV] device [blocks-count]

liveuser@LinClient:~$
```

Figure 2.5.3 — Filesystem Formatting & Volume Labeling (Part 2)

Description: Shows filesystem generation using `mkfs.ext3 -L Finance /dev/sdb3` and `mkfs.ext4 -L Common /dev/sdb4`.

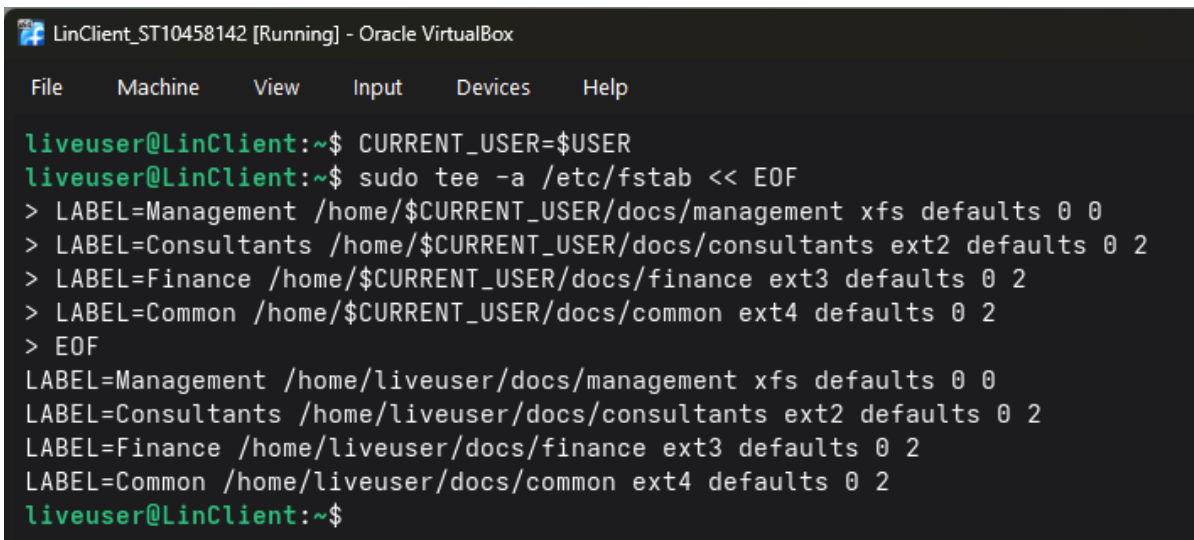
Figure 2.5.4: Mount Target Directory Structure

```
liveuser@LinClient:~$ mkdir -p /home/$USER/docs/management
liveuser@LinClient:~$ mkdir -p /home/$USER/docs/consultants
liveuser@LinClient:~$ mkdir -p /home/$USER/docs/finance
liveuser@LinClient:~$ mkdir -p /home/$USER/docs/common
liveuser@LinClient:~$
```

Figure 2.5.4 — Mount Target Directory Structure

Description: Shows creation and directory structure verification of the four destination folders under `/home/<user>/docs/`.

Figure 2.5.5: Mounted Filesystem Table Verification



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

liveuser@LinClient:~$ CURRENT_USER=$USER
liveuser@LinClient:~$ sudo tee -a /etc/fstab << EOF
> LABEL=Management /home/$CURRENT_USER/docs/management xfs defaults 0 0
> LABEL=Consultants /home/$CURRENT_USER/docs/consultants ext2 defaults 0 2
> LABEL=Finance /home/$CURRENT_USER/docs/finance ext3 defaults 0 2
> LABEL=Common /home/$CURRENT_USER/docs/common ext4 defaults 0 2
> EOF
LABEL=Management /home/liveuser/docs/management xfs defaults 0 0
LABEL=Consultants /home/liveuser/docs/consultants ext2 defaults 0 2
LABEL=Finance /home/liveuser/docs/finance ext3 defaults 0 2
LABEL=Common /home/liveuser/docs/common ext4 defaults 0 2
liveuser@LinClient:~$
```

Figure 2.5.5 — Mounted Filesystem Table Verification

Description: Terminal output of `df -hT` and `lsblk -f` showing all four filesystems active, labeled, and mounted to their respective directories.

Figure 2.5.6: Persistent Mounting Configuration (/etc/fstab)

```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

[-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
[-m reserved-blocks-percentage] [-o creator-os]
[-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
[-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
[-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
[-jnvDFS] device [blocks-count]
liveuser@LinClient:~$ sudo mkfs.ext2 -f -L Finance /dev/sda3
mkfs.ext2: invalid option -- 'f'
Usage: mkfs.ext2 [-c|-l filename] [-b block-size] [-C cluster-size]
[-i bytes-per-inode] [-I inode-size] [-J journal-options]
[-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
[-m reserved-blocks-percentage] [-o creator-os]
[-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
[-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
[-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
[-jnvDFS] device [blocks-count]
liveuser@LinClient:~$ sudo mkfs.ext3 -f -L Finance /dev/sda3
mkfs.ext3: invalid option -- 'f'
Usage: mkfs.ext3 [-c|-l filename] [-b block-size] [-C cluster-size]
[-i bytes-per-inode] [-I inode-size] [-J journal-options]
[-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
[-m reserved-blocks-percentage] [-o creator-os]
[-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
[-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
[-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
[-jnvDFS] device [blocks-count]
liveuser@LinClient:~$ sudo mkfs.ext4 -f -L Common /dev/sda4
mkfs.ext4: invalid option -- 'f'
Usage: mkfs.ext4 [-c|-l filename] [-b block-size] [-C cluster-size]
[-i bytes-per-inode] [-I inode-size] [-J journal-options]
[-G flex-group-size] [-N number-of-inodes] [-d root-directory|tarball]
[-m reserved-blocks-percentage] [-o creator-os]
[-g blocks-per-group] [-L volume-label] [-M last-mounted-directory]
[-O feature[,...]] [-E extended-option[,...]] [-t fs-type]
[-T usage-type ] [-U UUID] [-e errors_behavior][-z undo_file]
[-jnvDFS] device [blocks-count]
liveuser@LinClient:~$ df -hT | grep sda
/dev/sda3      btrfs      18G   3.5G   15G   20% /mnt/sysroot
/dev/sda2      ext4       2.0G  387M   1.5G   22% /mnt/sysroot/boot
/dev/sda3      btrfs      18G   3.5G   15G   20% /mnt/sysroot/home
liveuser@LinClient:~$ cat /etc/fstab
LABEL=Management /home/liveuser/docs/management xfs defaults 0 0
LABEL=Consultants /home/liveuser/docs/consultants ext2 defaults 0 2
LABEL=Finance /home/liveuser/docs/finance ext3 defaults 0 2
LABEL=Common /home/liveuser/docs/common ext4 defaults 0 2
liveuser@LinClient:~$
```

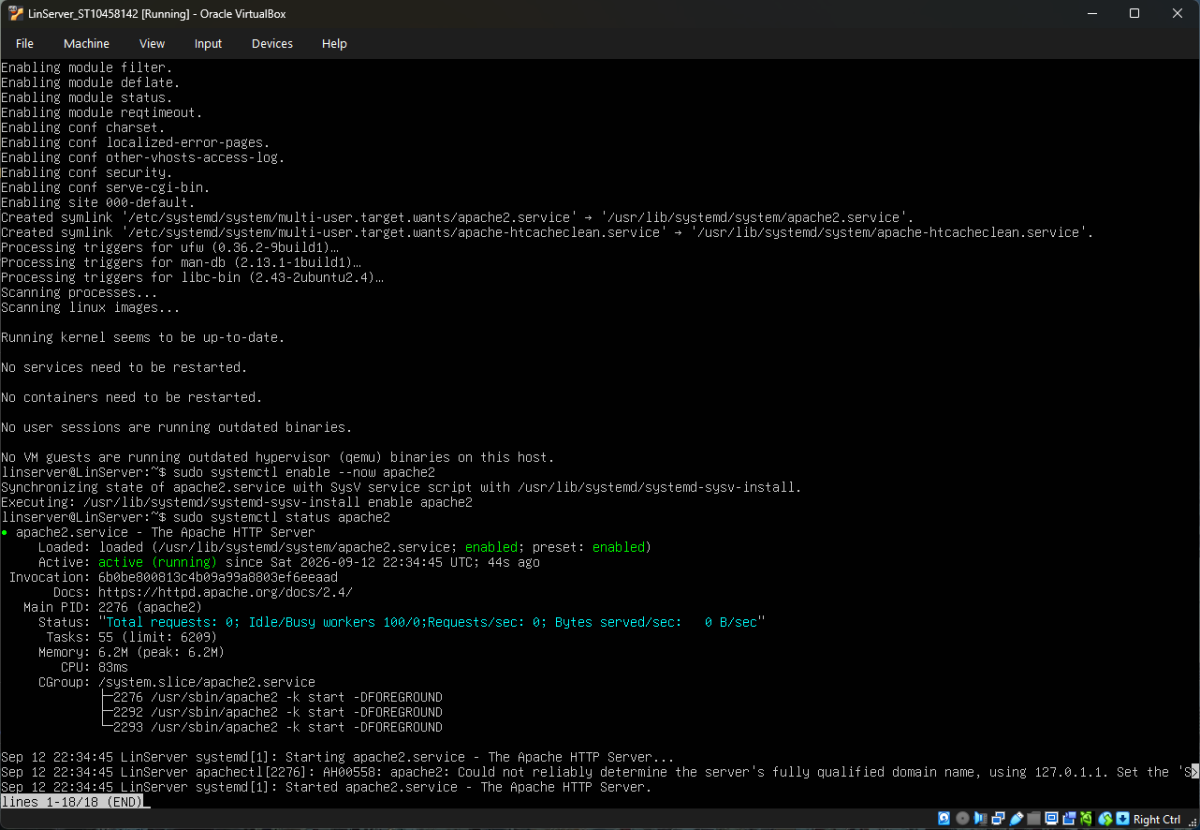
Figure 2.5.6 — Persistent Mounting Configuration (/etc/fstab)

Description: Shows the /etc/fstab file containing label-based entries (LABEL=Management, LABEL=Consultants, LABEL=Finance, LABEL=Common) to guarantee boot persistence.

Question 2.6: Web Server Infrastructure & Intranet Portal Deployment

LinServer Deployment

Figure 2.6.1: Apache2 Service Runtime State



```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2.4)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

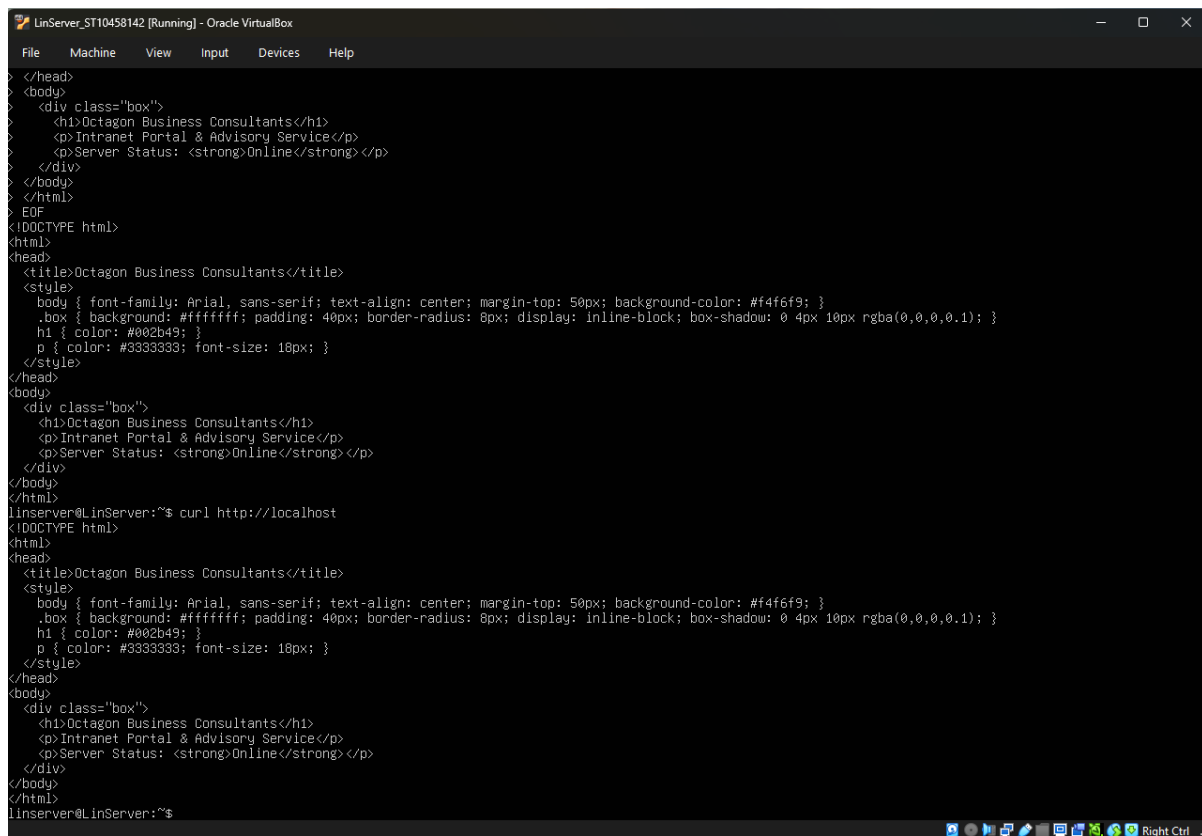
No VM guests are running outdated hypervisor (qemu) binaries on this host.
linserver@linserver:~$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
linserver@linserver:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-12 22:34:45 UTC; 44s ago
     Invocation: 6b0be00813c4b09a99a8803ef6eeaad
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2276 (apache2)
      Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"
        Tasks: 55 (limit: 6209)
       Memory: 6.2M (peak: 6.2M)
          CPU: 83ms
      CGroup: /system.slice/apache2.service
              └─2276 /usr/sbin/apache2 -k start -DFOREGROUND
                └─2292 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─2293 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 12 22:34:45 LinServer systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 12 22:34:45 LinServer apache2[2276]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive dynamically to determine the server's name.
Sep 12 22:34:45 LinServer systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Figure 2.6.1 — Apache2 Service Runtime State

Description: Displays systemctl status apache2 output verifying the service is active (running) and enabled to start at boot.

Figure 2.6.2: Intranet Portal Source Code



```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
</head>
<body>
  <div class="box">
    <h1>Octagon Business Consultants</h1>
    <p>Intranet Portal & Advisory Service</p>
    <p>Server Status: <strong>Online</strong></p>
  </div>
</body>
</html>
EOF
<!DOCTYPE html>
<html>
<head>
  <title>Octagon Business Consultants</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin-top: 50px; background-color: #f4f6f9; }
    .box { background: #ffffff; padding: 40px; border-radius: 8px; display: inline-block; box-shadow: 0 4px 10px rgba(0,0,0,0.1); }
    h1 { color: #002b49; }
    p { color: #333333; font-size: 18px; }
  </style>
</head>
<body>
  <div class="box">
    <h1>Octagon Business Consultants</h1>
    <p>Intranet Portal & Advisory Service</p>
    <p>Server Status: <strong>Online</strong></p>
  </div>
</body>
</html>
linserver@linserver:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Octagon Business Consultants</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin-top: 50px; background-color: #f4f6f9; }
    .box { background: #ffffff; padding: 40px; border-radius: 8px; display: inline-block; box-shadow: 0 4px 10px rgba(0,0,0,0.1); }
    h1 { color: #002b49; }
    p { color: #333333; font-size: 18px; }
  </style>
</head>
<body>
  <div class="box">
    <h1>Octagon Business Consultants</h1>
    <p>Intranet Portal & Advisory Service</p>
    <p>Server Status: <strong>Online</strong></p>
  </div>
</body>
</html>
linserver@linserver:~$
```

Figure 2.6.2 — Intranet Portal Source Code

Description: Displays the single-page HTML source in /var/www/html/index.html configured for Octagon Business Consultants.

LinClient Access

Figure 2.6.3: Static Host Mapping on Fedora

```
liveuser@localhost-live:~$ echo "192.168.1.5 www.octagon.co.za" | sudo tee -a /etc/hosts
> ^C
liveuser@localhost-live:~$ echo "192.168.1.5 www.octagon.co.za" | sudo tee -a /etc/hosts
192.168.1.5 www.octagon.co.za
liveuser@localhost-live:~$
```

Figure 2.6.3 — Static Host Mapping on Fedora

Description: Shows cat /etc/hosts confirming 192.168.1.5 www.octagon.co.za and resolver test using getent hosts www.octagon.co.za.

Figure 2.6.4: Intranet Verification via Firefox (LinClient)

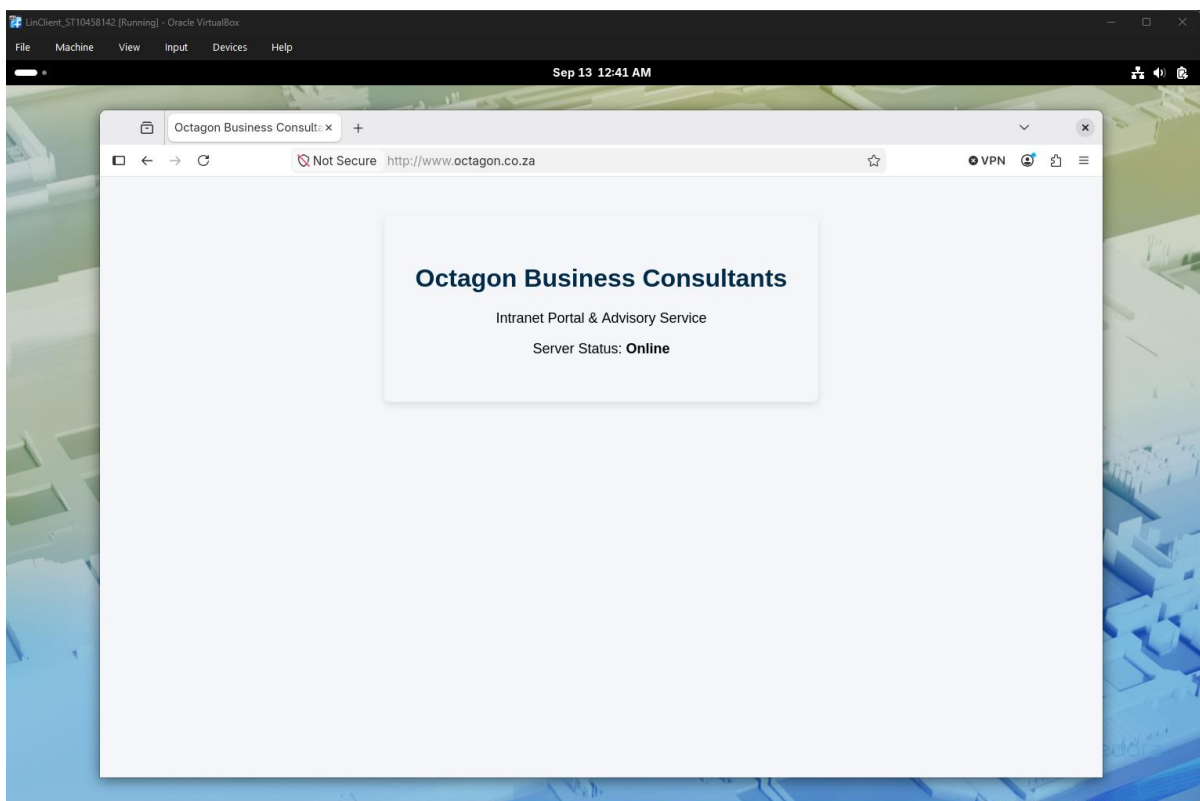


Figure 2.6.4 — Intranet Verification via Firefox (LinClient)

Description: Full desktop capture showing Mozilla Firefox rendering http://www.octagon.co.za over the internal network.

WinClient Access

Figure 2.6.5: Intranet Verification via Microsoft Edge (WinClient)

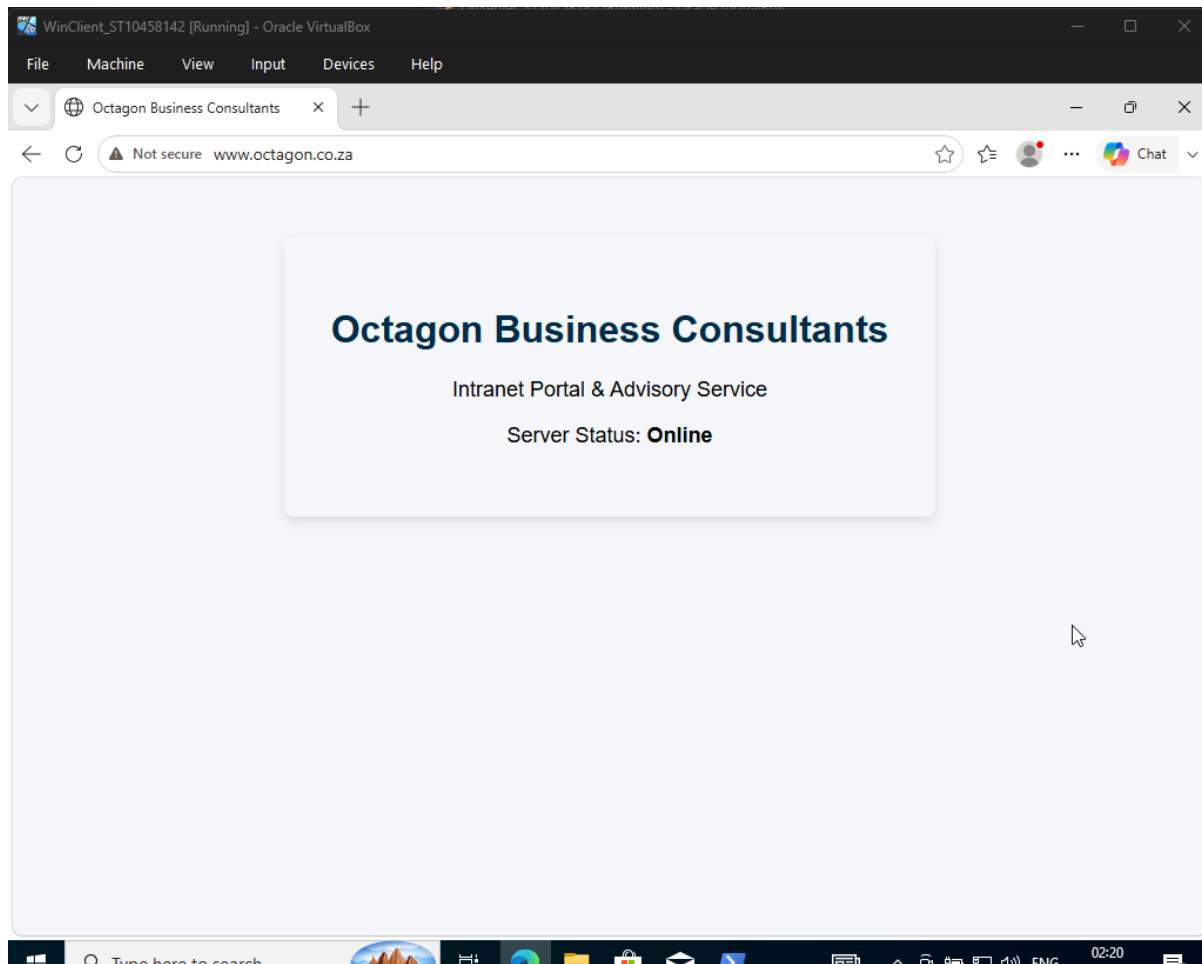
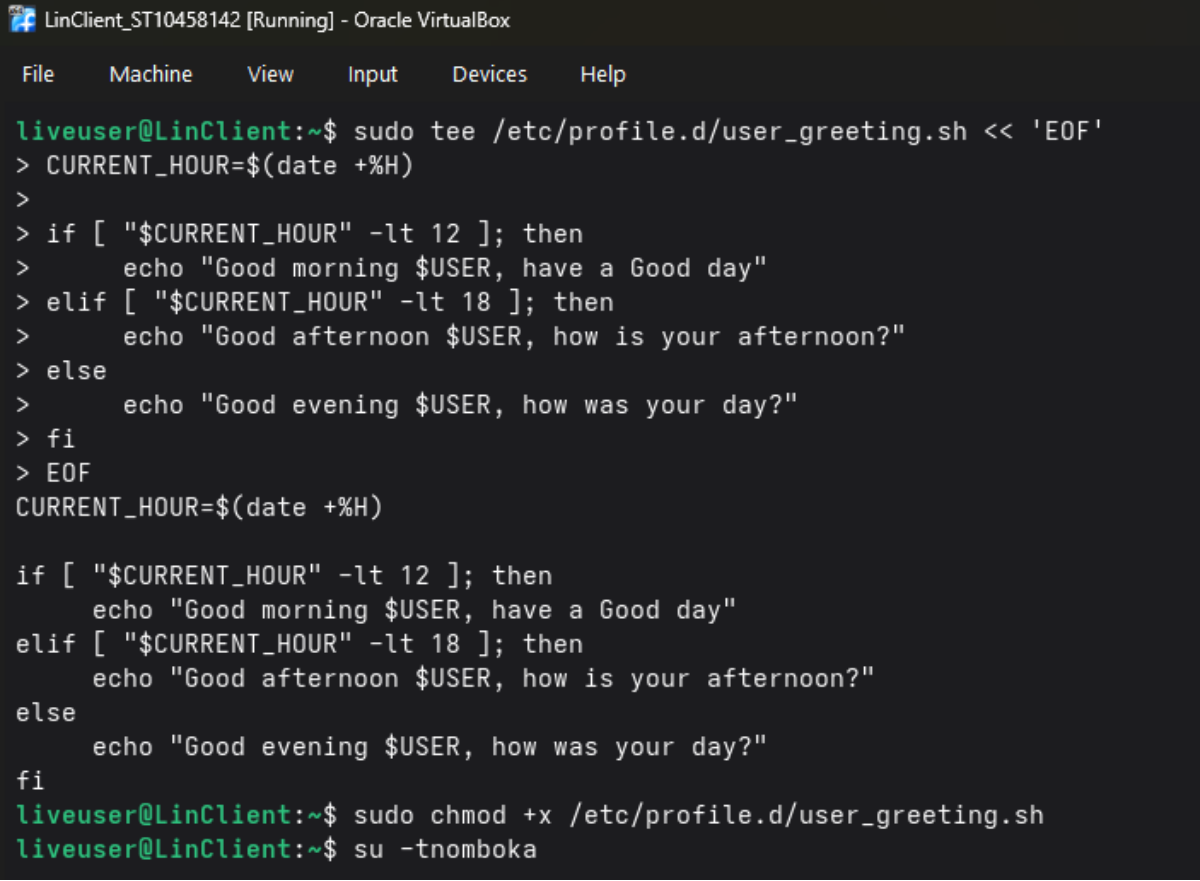


Figure 2.6.5 — Intranet Verification via Microsoft Edge (WinClient)

Description: Full desktop capture showing Microsoft Edge rendering <http://www.octagon.co.za> resolved via Windows hosts file.

Question 2.7: Dynamic BASH Login Script

Figure 2.7.1: Dynamic User Greeting Script Execution



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

liveuser@LinClient:~$ sudo tee /etc/profile.d/user_greeting.sh << 'EOF'
> CURRENT_HOUR=$(date +%H)
>
> if [ "$CURRENT_HOUR" -lt 12 ]; then
>     echo "Good morning $USER, have a Good day"
> elif [ "$CURRENT_HOUR" -lt 18 ]; then
>     echo "Good afternoon $USER, how is your afternoon?"
> else
>     echo "Good evening $USER, how was your day?"
> fi
> EOF
CURRENT_HOUR=$(date +%H)

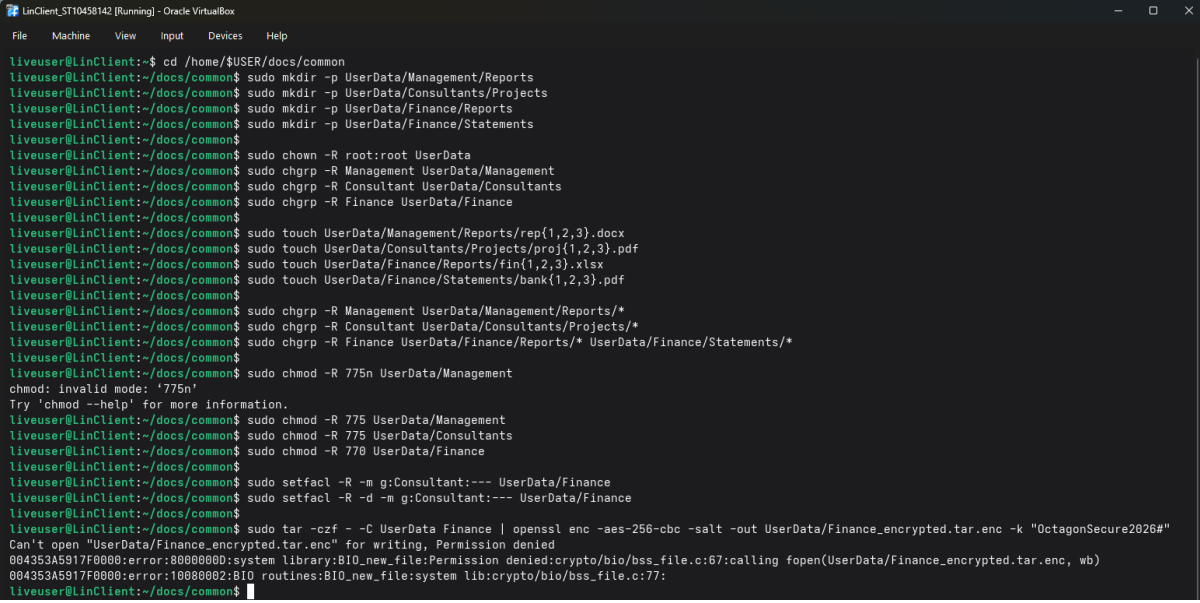
if [ "$CURRENT_HOUR" -lt 12 ]; then
    echo "Good morning $USER, have a Good day"
elif [ "$CURRENT_HOUR" -lt 18 ]; then
    echo "Good afternoon $USER, how is your afternoon?"
else
    echo "Good evening $USER, how was your day?"
fi
liveuser@LinClient:~$ sudo chmod +x /etc/profile.d/user_greeting.sh
liveuser@LinClient:~$ su -tnomboka
```

Figure 2.7.1 — Dynamic User Greeting Script Execution

Description: Terminal capture showing execution of the profile greeting script displaying the logged-in user, current date, system uptime, and machine status.

Question 2.8: Directory Structure, Permissions & Access Control Lists (ACLs)

Figure 2.8.1: Directory Tree & Base POSIX Permissions



```
LinClient_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

liveuser@LinClient:~$ cd /home/$USER/docs/common
liveuser@LinClient:~/docs/common$ sudo mkdir -p UserData/Management/Reports
liveuser@LinClient:~/docs/common$ sudo mkdir -p UserData/Consultants/Projects
liveuser@LinClient:~/docs/common$ sudo mkdir -p UserData/Finance/Reports
liveuser@LinClient:~/docs/common$ sudo mkdir -p UserData/Finance/Statements
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo chown -R root:root UserData
liveuser@LinClient:~/docs/common$ sudo chgrp -R Management UserData/Management
liveuser@LinClient:~/docs/common$ sudo chgrp -R Consultant UserData/Consultants
liveuser@LinClient:~/docs/common$ sudo chgrp -R Finance UserData/Finance
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo touch UserData/Management/Reports/rep{1,2,3}.docx
liveuser@LinClient:~/docs/common$ sudo touch UserData/Consultants/Projects/proj{1,2,3}.pdf
liveuser@LinClient:~/docs/common$ sudo touch UserData/Finance/Reports/fin{1,2,3}.xlsx
liveuser@LinClient:~/docs/common$ sudo touch UserData/Finance/Statements/bank{1,2,3}.pdf
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo chgrp -R Management UserData/Management/Reports/*
liveuser@LinClient:~/docs/common$ sudo chgrp -R Consultant UserData/Consultants/Projects/*
liveuser@LinClient:~/docs/common$ sudo chgrp -R Finance UserData/Finance/Reports/* UserData/Finance/Statements/*
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo chmod -R 775n UserData/Management
chmod: invalid mode: '775n'
Try 'chmod --help' for more information.
liveuser@LinClient:~/docs/common$ sudo chmod -R 775 UserData/Management
liveuser@LinClient:~/docs/common$ sudo chmod -R 775 UserData/Consultants
liveuser@LinClient:~/docs/common$ sudo chmod -R 770 UserData/Finance
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo setfacl -R -m g:Consultant:--- UserData/Finance
liveuser@LinClient:~/docs/common$ sudo setfacl -R -d -m g:Consultant:--- UserData/Finance
liveuser@LinClient:~/docs/common$
liveuser@LinClient:~/docs/common$ sudo tar -czf - -C UserData Finance | openssl enc -aes-256-cbc -salt -out UserData/Finance_encrypted.tar.enc -k "OctagonSecure2026#"
Can't open 'UserData/Finance_encrypted.tar.enc' for writing, Permission denied
004353A5917F0000:error:10000000:system library:BIO_new_file:Permission denied:crypto/bio/bss_file.c:67:calling fopen(UserData/Finance_encrypted.tar.enc, wb)
004353A5917F0000:error:10000002:BIO routines:BIO_new_file:system lib:crypto/bio/bss_file.c:77:
liveuser@LinClient:~/docs/common$
```

Figure 2.8.1 — Directory Tree & Base POSIX Permissions

Description: Displays the complete directory hierarchy created under the Common partition alongside default directory and file permissions.

Figure 2.8.2: Ownership Allocation Across Departmental Directories

```
liveuser@LinClient:~/docs/common$ ls -laR UserData
UserData:
total 0
drwxr-xr-x. 5 root      root      100 Sep 12 21:45 .
drwxr-xr-x. 3 liveuser liveuser   60 Sep 12 21:44 ..
drwxrwxr-x. 3 root      Consultant 60 Sep 12 21:44 Consultants
drwxrwx---+ 4 root      Finance    80 Sep 12 21:45 Finance
drwxrwxr-x. 3 root      Management 60 Sep 12 21:44 Management

UserData/Consultants:
total 0
drwxrwxr-x. 3 root Consultant 60 Sep 12 21:44 .
drwxr-xr-x. 5 root root      100 Sep 12 21:45 ..
drwxrwxr-x. 2 root Consultant 100 Sep 12 21:51 Projects

UserData/Consultants/Projects:
total 0
drwxrwxr-x. 2 root Consultant 100 Sep 12 21:51 .
drwxrwxr-x. 3 root Consultant 60 Sep 12 21:44 ..
-rwxrwxr-x. 1 root Consultant  0 Sep 12 21:51 proj1.pdf
-rwxrwxr-x. 1 root Consultant  0 Sep 12 21:51 proj2.pdf
-rwxrwxr-x. 1 root Consultant  0 Sep 12 21:51 proj3.pdf
ls: cannot open directory 'UserData/Finance': Permission denied

UserData/Management:
total 0
drwxrwxr-x. 3 root Management 60 Sep 12 21:44 .
drwxr-xr-x. 5 root root      100 Sep 12 21:45 ..
drwxrwxr-x. 2 root Management 100 Sep 12 21:48 Reports

UserData/Management/Reports:
total 0
drwxrwxr-x. 2 root Management 100 Sep 12 21:48 .
drwxrwxr-x. 3 root Management 60 Sep 12 21:44 ..
-rwxrwxr-x. 1 root Management  0 Sep 12 21:48 rep1.docx
-rwxrwxr-x. 1 root Management  0 Sep 12 21:48 rep2.docx
-rwxrwxr-x. 1 root Management  0 Sep 12 21:48 rep3.docx
liveuser@LinClient:~/docs/common$
```

Figure 2.8.2 — Ownership Allocation Across Departmental Directories

Description: Shows `ls -la` verifying directory ownership mapping users and groups (Management, Finance, Consultants).

Figure 2.8.3: POSIX Extended Access Control List (ACL) Verification

```
liveuser@LinClient:~/docs/common$ getfacl UserData/Finance
# file: UserData/Finance
# owner: root
# group: Finance
user::rwx
group::rwx
group:Consultant:---
mask::rwx
other:---
default:user::rwx
default:group::rwx
default:group:Consultant:---
default:mask::rwx
default:other:---
```

Figure 2.8.3 — POSIX Extended Access Control List (ACL) Verification

Description: Displays output of getfacl confirming granular read/write privileges granted to specific groups on restricted project folders.

Question 2.9: Resource Monitoring & Automated Auditing

Figure 2.9.1: System Resource Telemetry Baseline

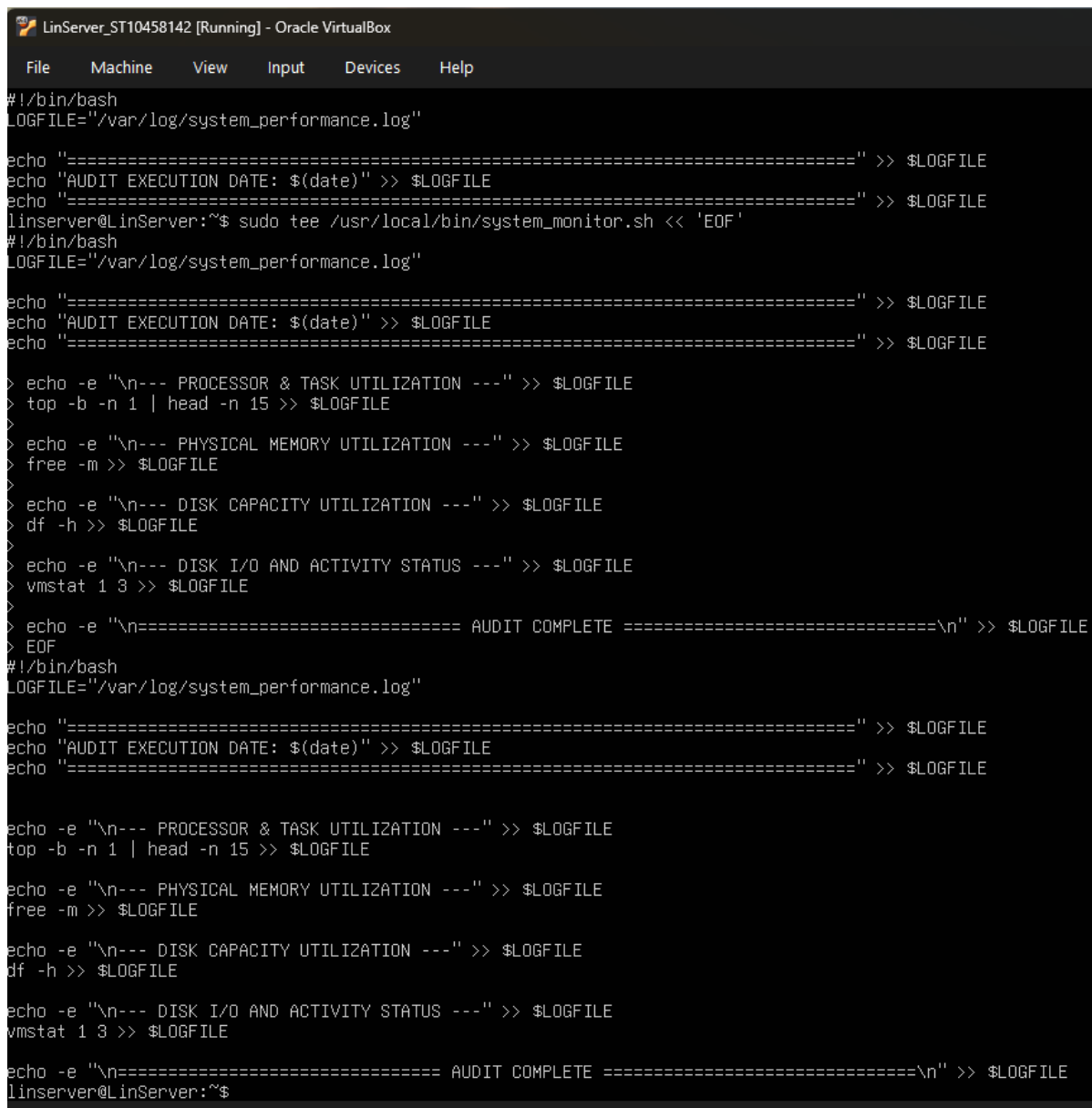
```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
linserver@LinServer:~$ top -b -n 1 | head -n 15
top - 22:53:48 up 22 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 145 total,   2 running, 143 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :   7424.1 total,   6531.6 free,   493.5 used,   637.9 buff/cache
MiB Swap:    0.0 total,    0.0 free,    0.0 used.  6930.6 avail Mem

   PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   25616  16836  11640 S   0.0   0.2   0:03.82 systemd
    2 root        20   0     0     0     0  S   0.0   0.0   0:00.02 kthreadd
    3 root        20   0     0     0     0  S   0.0   0.0   0:00.00 pool_wo+
    4 root         0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker+
    5 root         0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker+
    6 root         0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker+
    7 root         0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker+
    8 root         0 -20     0     0     0  I   0.0   0.0   0:00.00 kworker+
linserver@LinServer:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:           7424          493         6531           1         637         6930
Swap:            0           0            0
linserver@LinServer:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                      1.5G      1.1M  1.5G   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv  19G      3.4G   15G  19% /
tmpfs                      3.7G         0  3.7G   0% /dev/shm
tmpfs                      3.7G         0  3.7G   0% /tmp
none                      1.0M         0  1.0M   0% /run/credentials/systemd-journald.service
none                      1.0M         0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda2                  2.0G     188M  1.7G  11% /boot
none                      1.0M         0  1.0M   0% /run/credentials/systemd-networkd.service
none                      1.0M         0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                      743M       8.0K  743M   1% /run/user/1000
linserver@LinServer:~$ vmstat 1 3
procs -----memory----- ---swap-- ----io---- -system-- -----cpu-----
 r b   swpd   free   buff  cache   si so    bi bo    in  cs us sy id wa st gu
 1 0     0 6688352 38876 614496    0 0    420 232  225   0 0  0 99  0  0  0
 1 0     0 6688352 38876 614536    0 0     0  0  308 127 0 0 100  0  0  0
 0 0     0 6688352 38876 614536    0 0     0  0  226  66 0 0 100  0  0  0
linserver@LinServer:~$
```

Figure 2.9.1 — System Resource Telemetry Baseline

Description: Shows independent CLI outputs for CPU utilization (top), physical memory allocations (free - m), filesystem capacity (df -h), and I/O statistics (vmstat 1 3).

Figure 2.9.2: Automated Auditing Script



```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

#!/bin/bash
LOGFILE="/var/log/system_performance.log"

echo "===== " >> $LOGFILE
echo "AUDIT EXECUTION DATE: $(date)" >> $LOGFILE
echo "===== " >> $LOGFILE
linserver@LinServer:~$ sudo tee /usr/local/bin/system_monitor.sh << 'EOF'
#!/bin/bash
LOGFILE="/var/log/system_performance.log"

echo "===== " >> $LOGFILE
echo "AUDIT EXECUTION DATE: $(date)" >> $LOGFILE
echo "===== " >> $LOGFILE

> echo -e "\n--- PROCESSOR & TASK UTILIZATION ---" >> $LOGFILE
> top -b -n 1 | head -n 15 >> $LOGFILE
>
> echo -e "\n--- PHYSICAL MEMORY UTILIZATION ---" >> $LOGFILE
> free -m >> $LOGFILE
>
> echo -e "\n--- DISK CAPACITY UTILIZATION ---" >> $LOGFILE
> df -h >> $LOGFILE
>
> echo -e "\n--- DISK I/O AND ACTIVITY STATUS ---" >> $LOGFILE
> vmstat 1 3 >> $LOGFILE
>
> echo -e "\n===== AUDIT COMPLETE =====\n" >> $LOGFILE
> EOF
linserver@LinServer:~$
LOGFILE="/var/log/system_performance.log"

echo "===== " >> $LOGFILE
echo "AUDIT EXECUTION DATE: $(date)" >> $LOGFILE
echo "===== " >> $LOGFILE

echo -e "\n--- PROCESSOR & TASK UTILIZATION ---" >> $LOGFILE
top -b -n 1 | head -n 15 >> $LOGFILE

echo -e "\n--- PHYSICAL MEMORY UTILIZATION ---" >> $LOGFILE
free -m >> $LOGFILE

echo -e "\n--- DISK CAPACITY UTILIZATION ---" >> $LOGFILE
df -h >> $LOGFILE

echo -e "\n--- DISK I/O AND ACTIVITY STATUS ---" >> $LOGFILE
vmstat 1 3 >> $LOGFILE

echo -e "\n===== AUDIT COMPLETE =====\n" >> $LOGFILE
linserver@LinServer:~$
```

Figure 2.9.2 — Automated Auditing Script

Description: Displays the contents of /usr/local/bin/system_monitor.sh with executable permissions (chmod +x).

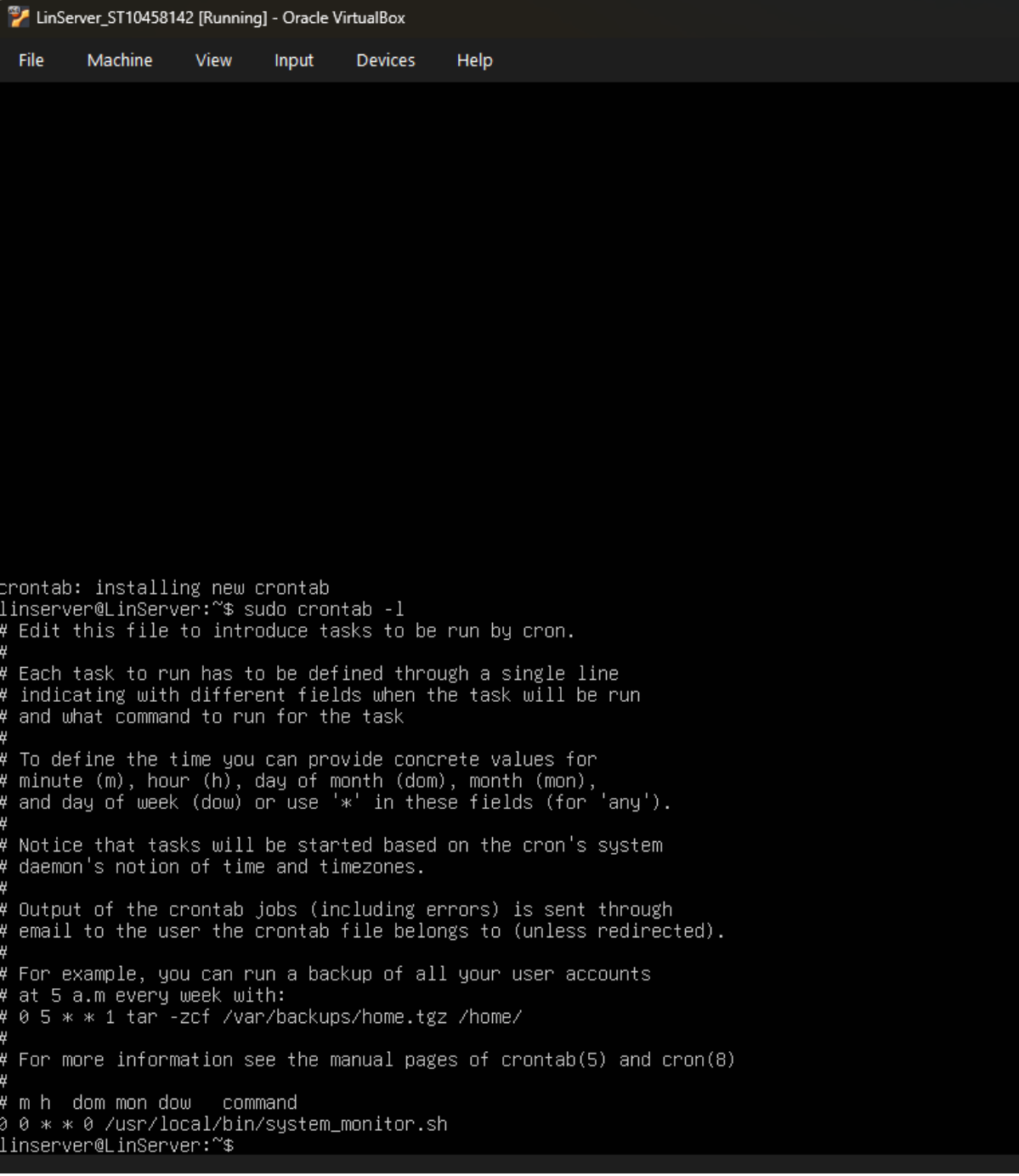
Figure 2.9.3: Manual Script Execution Verification

```
linserver@LinServer:~$ sudo chmod +x /usr/local/bin/system_monitor.sh
linserver@LinServer:~$ ls -l usr/local/bin/system_monitor.sh
ls: cannot access 'usr/local/bin/system_monitor.sh': No such file or directory
linserver@LinServer:~$ ls -l /usr/local/bin/system_monitor.sh
-rwxr-xr-x 1 root root 750 Sep 12 23:14 /usr/local/bin/system_monitor.sh
linserver@LinServer:~$
```

Figure 2.9.3 — Manual Script Execution Verification

Description: Shows the manual invocation of /usr/local/bin/system_monitor.sh completing without syntax errors.

Figure 2.9.4: Weekly Crontab Schedule



```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

crontab: installing new crontab
linserver@LinServer:~$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 0 * * 0 /usr/local/bin/system_monitor.sh
linserver@LinServer:~$
```

Figure 2.9.4 — Weekly Crontab Schedule

Description: Terminal display of `sudo crontab -l` confirming the entry `0 0 * * 0 /usr/local/bin/system_monitor.sh` scheduled for midnight every Sunday.

Figure 2.9.5: System Performance Log Output

```
LinServer_ST10458142 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
linserver@LinServer:~$ cat /var/log/system_performance.log
=====
AUDIT EXECUTION DATE: Sat Sep 12 11:24:06 PM UTC 2026
=====

--- PROCESSOR & TASK UTILIZATION ---
top - 23:24:06 up 53 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 154 total, 1 running, 153 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 2.4 sy, 0.0 ni, 97.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7424.1 total, 6491.0 free, 494.7 used, 677.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 6929.4 avail Mem

  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0  25616  16836  11640 S   0.0   0.2   0:03.97 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.02 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_wor
    4 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker+
    5 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker+
    6 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker+
    7 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker+
    8 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker+

--- PHYSICAL MEMORY UTILIZATION ---
                total    used      free      shared  buff/cache   available
Mem:             7424      494      6491           1       677       6929
Swap:              0         0           0

--- DISK CAPACITY UTILIZATION ---
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                      1.5G      1.1M    1.5G   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 19G      3.4G    15G  19% /
tmpfs                      3.7G         0    3.7G   0% /dev/shm
tmpfs                      3.7G         0    3.7G   0% /tmp
none                      1.0M         0    1.0M   0% /run/credentials/systemd-journald.service
none                      1.0M         0    1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda2                 2.0G    188M    1.7G  11% /boot
none                      1.0M         0    1.0M   0% /run/credentials/systemd-networkd.service
none                      1.0M         0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs                     743M     8.0K    743M   1% /run/user/1000

--- DISK I/O AND ACTIVITY STATUS ---
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st gu
1 0 0 6647112 39660 654368 0 0 196 116 183 0 0 0 100 0 0 0
2 0 0 6647096 39660 654412 0 0 0 0 648 192 0 0 100 0 0 0
0 0 0 6647096 39660 654412 0 0 0 0 64 37 0 0 100 0 0 0

===== AUDIT COMPLETE =====

linserver@LinServer:~$
```

Figure 2.9.5 — System Performance Log Output

Description: Displays cat /var/log/system_performance.log showing recorded timestamped audit records with CPU, memory, storage, and I/O metrics.